Journal of Computational and Applied Mathematics 452 (2024) 116115

Contents lists available at ScienceDirect

Journal of Computational and Applied Mathematics

journal homepage: www.elsevier.com/locate/cam

The mixed method with two Lagrange multiplier formulations for the Signorini problem

Qi Wang [a], Jingyan Hu [a], Guanyu Zhou [a,b,*,1]

[a] Institute of Fundamental and Frontier Sciences, University of Electronic Science and Technology of China, Chengdu, 610051, China
[b] School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu, 610051, China

ARTICLE INFO

Keywords:
Signorini condition
Mixed finite element method
Active/Inactive set
Projection method

ABSTRACT

Introducing stress as a new unknown, we consider the Signorini problem in the mixed form. The well-posedness theory of the continuous and discrete mixed variational inequalities has been established by reformulating them into projection problems. Two Lagrange multiplier formulations are introduced utilizing different projections. The equivalency among the Lagrange multiplier formulations and the mixed variational inequality is demonstrated in continuous and discrete sense, respectively. For the discrete mixed variational inequality, we develop the error estimates. Based on two Lagrange multiplier formulations, we design two Active/Inactive set algorithms and investigate the convergence. Several numerical experiments are conducted to verify the theoretical convergence rates of the finite element discretization and the iteration algorithms.

1. Introduction

The Signorini problem was first introduced by Signorini in [1] to simulate the interaction between a linearly elastic body and a rigid foundation without friction. This problem has gained significant interest among engineers and mathematicians ever since and has been applied to various fields, including electropainting [2,3], the elastic contact problem [4–8], and the artificial outflow boundary condition of viscous flow [9–11].

In this discussion, we will focus on the simplified Signorini problem and present a theoretical analysis of our proposed ideas. These findings can be applied to treat the generalized Signorini problem in elastic mechanics without any challenges.

Let Ω be a smooth and bounded domain in $\mathbb{R}^d$ $(d = 2, 3)$. Let $\boldsymbol{n}$ denote the unit outer normal vector to the boundary $\partial\Omega$, where $\partial\Omega = \overline{\Gamma_D \cup \Gamma_S}$, $\Gamma_D \cap \Gamma_S = \emptyset$, $|\Gamma_D| > 0$ and $|\Gamma_S| > 0$. The Signorini problem **(P)** is presented as follows:

$$-\Delta u = f \qquad \text{in } \Omega, \tag{1a}$$

$$u = 0 \qquad \text{on } \Gamma_D, \tag{1b}$$

$$u \geq 0, \quad \nabla u \cdot \boldsymbol{n} \geq 0, \quad u(\nabla u \cdot \boldsymbol{n}) = 0 \qquad \text{on } \Gamma_S, \tag{1c}$$

where $f \in L^2(\Omega)$ is a given function. To introduce the weak form of **(P)**, we set the function spaces:

$$X = \{v \in H^1(\Omega) : v|_{\Gamma_D} = 0\}, \quad \tilde{X} := \{v \in X : v|_{\Gamma_S} \geq 0\},$$

* Corresponding author.
E-mail addresses: qi_wang@std.uestc.edu.cn (Q. Wang), jingyan_hu@std.uestc.edu.cn (J. Hu), wind_geno@live.com, zhoug@uestc.edu.cn (G. Zhou).

[1] The research of Guanyu Zhou was partially supported by NSFC General Project (No. 12171071) and the Natural Science Foundation of Sichuan Province (No. 2023NSFSC0055).

https://doi.org/10.1016/j.cam.2024.116115
Received 23 October 2023; Received in revised form 24 May 2024
Available online 28 June 2024
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$\tilde{\Lambda}_{-1/2} := \{\mu \in H^{-\frac{1}{2}}(\Gamma_S) : \mu \geq 0\}$(see Remark 2.2 for the meaning of $\mu \geq 0$).

Conventionally, the problem is formulated into the variational inequality of the first kind, stated as follows.

(VI) Find $u \in \tilde{X}$ such that

$$(\nabla u, \nabla(v-u)) \geq (f, v-u) \qquad \forall v \in \tilde{X},$$

where $(\cdot,\cdot)$ represents the inner product of $L^2(\Omega)$. Setting $\lambda = \nabla u \cdot \boldsymbol{n}$, such variational inequality can be reformulated into the equivalent Lagrange multiplier formulation:

(VL) Find $(u, \lambda) \in X \times \tilde{\Lambda}_{-1/2}$, such that

$$\begin{aligned} &(\nabla u, \nabla v) - \langle \lambda, v\rangle_{\Gamma_S} = (f, v) && \forall v \in X, \\ &\langle \mu - \lambda, u\rangle_{\Gamma_S} \geq 0 && \forall \mu \in \tilde{\Lambda}_{-1/2}. \end{aligned}$$

where $\langle a, b\rangle_{\Gamma_S} := \int_{\Gamma_S} ab \ \ d\Gamma$ for all $a \in \Lambda_{-1/2}$ and $b \in \Lambda_{1/2}$.

There are many numerical methods available for solving **(VI)** (or **(VL)**), including the finite element method (FEM) (see [7,12–17], just to name a few), the boundary element method (BEM) [18–20], the discontinuous Galerkin method and the discontinuous Petrov–Galerkin method (for the reaction-dominated case) [21,22], and so on. Furthermore, several iteration algorithms have been developed to compute the numerical solution of the variational inequality. These include the Uzawa method [15], the Active/Inactive set method (also known as the Primal-dual Active set method) [23,24], the switching algorithm [3], and the over-relaxation method [15]. In addition, the penalty method [6,15,25] is also a popular approach to solving the problem.

Note that, in some literature [4,25–29], the discretization of **(VL)** is called the mixed variational inequality. To be clear, the mixed form considered in this paper takes $\boldsymbol{\sigma} := \nabla u$ as an additional unknown. That is, instead of **(P)**, we consider the weak formulations and numerical discretization of the following formally equivalent system **(MP)**:

$$\begin{aligned} \boldsymbol{\sigma} - \nabla u &= 0 && \text{in } \Omega, && (3a)\\ \nabla \cdot \boldsymbol{\sigma} + f &= 0 && \text{in } \Omega, && (3b)\\ u &= 0 && \text{on } \Gamma_D, && (3c)\\ u \geq 0, \quad \boldsymbol{\sigma}\cdot\boldsymbol{n} \geq 0, \quad u\boldsymbol{\sigma}\cdot\boldsymbol{n} &= 0 && \text{on } \Gamma_S. && (3d) \end{aligned}$$

To write the variational form of **(MP)**, we introduce the function spaces:

$$V := H(\mathrm{div};\Omega), \quad \tilde{V} := \{\boldsymbol{\tau} \in V : \boldsymbol{\tau}\cdot\boldsymbol{n} \geq 0 \text{ on } \Gamma_S\} \quad (\text{ see Remark 2.2for the definition of } \boldsymbol{\tau}\cdot\boldsymbol{n} \geq 0),$$

$$Q := L^2(\Omega), \quad \Lambda_{1/2} := H_{00}^{\frac{1}{2}}(\Gamma_S), \quad \tilde{\Lambda}_{1/2} := \{p \in \Lambda_{1/2} : p \geq 0 \text{ a.e.}\}, \quad \Lambda_{-1/2} := H^{-\frac{1}{2}}(\Gamma_S) = \Lambda_{1/2}^*.$$

Testing (3a) and (3b) by $\boldsymbol{\tau} \in \tilde{V}$ and $v \in Q$ respectively, and using the integration by parts, one can reformulate **(MP)** into the following mixed variational inequality.

(MVI) Find $(\boldsymbol{\sigma}, u) \in \tilde{V} \times Q$ such that

$$\begin{aligned} (\boldsymbol{\sigma}, \boldsymbol{\tau}-\boldsymbol{\sigma}) + (u, \nabla\cdot(\boldsymbol{\tau}-\boldsymbol{\sigma})) &\geq 0 && \forall \boldsymbol{\tau} \in \tilde{V}, && (4a)\\ (\nabla\cdot\boldsymbol{\sigma}, v) + (f, v) &= 0 && \forall v \in Q. && (4b) \end{aligned}$$

An advantage of **(MVI)** is that the introduced stress $\boldsymbol{\sigma}$ can be approximated directly, which avoids additional errors from numerical post-processing. In comparison to **(VL)**, the regularity of the unknown u in **(MVI)** is lower, which allows for choosing more finite element subspaces, such as the P0-element. Moreover, the Dirichlet boundary condition can be naturally incorporated into the continuous and discrete formulations in mixed form (see for a more detailed discussion [30]). In addition, we will illustrate that, for the Signorini problem, designing iteration schemes becomes more flexible when considering the mixed formulation.

The theories of abstract mixed variational inequalities were first established by [27]. [31] and [5] investigated mixed methods for the elastoplasticity problem, with the latter studying the friction-type boundary and providing a regularized approach for approximation. In [8], a mixed formulation with the penalty-type stabilization term $\frac{1}{\varepsilon}(\nabla\cdot\boldsymbol{\sigma}_\varepsilon + f)$ is proposed for the unilateral elastic contact problem. The mixed finite element method for the Signorini problem has been further analyzed by [32–34], where [34] presents an Uzawa-type iteration scheme based on the minimization problem (the projection step is similar to Algorithm 4 of this work), and [32] gives a modified Uzawa algorithm and adopts the boundary element method to achieve the projection based on the discrete $H^{\frac{1}{2}}$-inner product in computation.

The main difference between our work and [32] is the treatment of the dual product $\langle \boldsymbol{\sigma}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S}$ in the discrete sense. [32] uses a projection formulation similar to the discrete version of (28) involving the discrete $H^{\frac{1}{2}}$-inner product on Γ_S and an operator from $H(\mathrm{div};\Omega)$ to the discrete $H^{\frac{1}{2}}(\Gamma_S)$ space. In our discrete problem, $\langle \boldsymbol{\sigma}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S}$ is discretized by the discrete L^2 inner product $(\boldsymbol{\sigma}_h\cdot\boldsymbol{n}, p_h)_{\Gamma_S}$ with P0-element on Γ_S. So the implementation of our scheme is simpler than that of [32]. We also obtain the well-posedness, stability, and error estimates. However, for the Lagrange multiplier p, we obtain the error estimate in L^2-norm, while [32] studies the error in the discrete $H^{\frac{1}{2}}$-norm. In computation, the L^2 projection is easier to carry out, and our algorithms also yield good convergence behavior.

It is worth noting that the Stokes/Navier–Stokes equations when accompanied by subgradient type boundary conditions [35–38], lead to the mixed variational (hemi-)inequalities with the velocity and pressure as the unknowns. If we introduce the stress as an additional unknown, the so-called fully-mixed variational (hemi-)inequalities are obtained.

The main results of this paper are summarized as follows. Firstly, we examine the well-posedness of the mixed variational formulations in both the continuous and discrete sense, where we use the RT0/P0-element to discretize $\boldsymbol{\sigma}$ and u. The unique existence of the mixed variational inequality has been shown by [32], which utilizes the coupled inf-sup condition involving the discrete $H^{\frac{1}{2}}(\Gamma_S)$-norm. In this work, we revisit the well-posedness of the continuous and discrete mixed variational inequalities and provide an alternative proof indicating that the mixed variational inequality is essentially a projection problem. Our analysis not involve the discrete $H^{\frac{1}{2}}(\Gamma_S)$-norm and the coupled inf-sup condition tools.

Secondly, we investigate two mixed Lagrange multiplier formulations for continuous and discrete cases. The continuous case is presented below.

(MLM-I) Find $(\boldsymbol{\sigma}, u, p) \in \tilde{V} \times Q \times \Lambda_{1/2}$ such that

$$(\boldsymbol{\sigma}, \boldsymbol{\tau}) + (u, \nabla \cdot \boldsymbol{\tau}) - \langle \boldsymbol{\tau} \cdot \boldsymbol{n}, p \rangle_{\Gamma_S} = 0 \qquad \forall \boldsymbol{\tau} \in V, \tag{5a}$$

$$(\nabla \cdot \boldsymbol{\sigma}, v) + (f, v) = 0 \qquad \forall v \in Q, \tag{5b}$$

$$\langle \boldsymbol{\tau} \cdot \boldsymbol{n} - \boldsymbol{\sigma} \cdot \boldsymbol{n}, p \rangle_{\Gamma_S} \geq 0 \qquad \forall \boldsymbol{\tau} \in \tilde{V}; \tag{5c}$$

(MLM-II) Find $(\boldsymbol{\sigma}, u, p) \in V \times Q \times \tilde{\Lambda}_{1/2}$ such that

$$(\boldsymbol{\sigma}, \boldsymbol{\tau}) + (u, \nabla \cdot \boldsymbol{\tau}) - \langle \boldsymbol{\tau} \cdot \boldsymbol{n}, p \rangle_{\Gamma_S} = 0 \qquad \forall \boldsymbol{\tau} \in V, \tag{6a}$$

$$(\nabla \cdot \boldsymbol{\sigma}, v) + (f, v) = 0 \qquad \forall v \in Q, \tag{6b}$$

$$\langle \boldsymbol{\sigma} \cdot \boldsymbol{n}, q - p \rangle_{\Gamma_S} \geq 0 \qquad \forall q \in \tilde{\Lambda}_{1/2}. \tag{6c}$$

If $u \in H^1(\Omega)$, then p equals the trace of u on Γ_S. See (36)(37) for the discrete case.

We demonstrate that the two Lagrange multiplier problems are equivalent. However, the implementation of these two Lagrange multiplier problems is carried out in quite different manners by using different projections. Roughly speaking, one ensures $\boldsymbol{\sigma} \cdot \boldsymbol{n} \geq 0$ on Γ_S by using the projection

$$\boldsymbol{\sigma} \cdot \boldsymbol{n} = \max(0, \boldsymbol{\sigma} \cdot \boldsymbol{n} - \rho p),$$

while the other enforce the constraint $u \geq 0$ on Γ_S by applying the projection

$$p = \max(0, p - \rho \boldsymbol{\sigma} \cdot \boldsymbol{n}).$$

Here, ρ is a positive parameter and p is the Lagrange multiplier of the mixed problem. In fact, $p = u|_{\Gamma_S}$ if $u \in H^1(\Omega)$.

Thirdly, we study the error analysis of the finite element discretization. Moreover, we design Active/Inactive set algorithms (Algorithms 1 and 2) based on two discrete mixed Lagrange multiplier formulations and investigate their convergence. We also provide two Uzawa algorithms in Appendix using the above two projections and analyze the convergence.

The rest of this paper is organized as follows. In Section 2, we introduce the mixed variational inequality and its two equivalent mixed multiplier formulations while demonstrating their well-posedness and equivalence. Section 3 is devoted to studying discrete mixed variational inequalities, discrete mixed Lagrange multiplier problems and Active/Inactive algorithms. Several numerical examples are provided in Section 4.

2. The mixed formulations

In this section, we revisit the well-posedness of the mixed variational inequality and show that **(MVL-I)** and **(MVL-II)** are both equivalent to **(MVI)**. Let us introduce the function spaces

$$\mathring{V} := \{\boldsymbol{\tau} \in V : \nabla \cdot \boldsymbol{\tau} = 0\}, \quad \mathring{\tilde{V}} := \{\mathring{\boldsymbol{\tau}} \in \mathring{V} : \mathring{\boldsymbol{\tau}} \cdot \boldsymbol{n} \geq 0 \text{ on } \Gamma_S\}$$

$$V_{\Gamma_S} := \{\boldsymbol{\tau} \in V : \boldsymbol{\tau} \cdot \boldsymbol{n} = 0 \text{ on } \Gamma_S\}, \quad \hat{V} := \{\boldsymbol{\tau} \in V : \boldsymbol{\tau} \cdot \boldsymbol{n} \in \Lambda\}$$

$$\Lambda := L^2(\Gamma_S) = \Lambda^*, \quad \tilde{\Lambda} := \{p \in \Lambda : p \geq 0 \text{ a.e.}\}.$$

Remark 2.1. By the extension theorem (or called the reverse trace inequality), for any $q \in \Lambda_{1/2}$, there exists a $v_q \in X$ satisfying $v_q|_{\Gamma_S} = q$ and

$$\|v_q\|_X \leq C_{E1} \|q\|_{\Lambda_{1/2}}, \tag{7}$$

where $C_{E1} > 0$ is a constant. For any $\boldsymbol{\tau} \in V$, the trace $\boldsymbol{\tau} \cdot \boldsymbol{n} \in \Lambda_{-1/2}$ is defined by

$$\langle \boldsymbol{\tau} \cdot \boldsymbol{n}, q \rangle_{\Gamma_S} = (\boldsymbol{\tau}, \nabla v_q) + (\nabla \cdot \boldsymbol{\tau}, v_q) \quad \forall q \in \Lambda_{1/2} \text{ and } v_q \in X, \tag{8}$$

which implies that (by using (7), (8) and $v|_{\Gamma_S} \in \Lambda_{1/2}$ for all $v \in X$)

$$\|\boldsymbol{\tau} \cdot \boldsymbol{n}\|_{\Lambda_{-1/2}} := \sup_{q \in \Lambda_{1/2}} \frac{\langle \boldsymbol{\tau} \cdot \boldsymbol{n}, q \rangle_{\Gamma_S}}{\|q\|_{\Lambda_{1/2}}} \leq \sup_{v_q \in X} \frac{(\boldsymbol{\tau}, \nabla v_q) + (\nabla \cdot \boldsymbol{\tau}, v_q)}{C_{E1}^{-1} \|v_q\|_X} \leq C_{E1} \|\boldsymbol{\tau}\|_V.$$

Remark 2.2 (*[39, Brézis]*). For any $\mu \in (C_0^\infty(\Gamma_S))^*$, we set

$$M(\mu) := \Big\{ a \in \mathbb{R} : \int_{\Gamma_S} \mu\phi \, ds \le a \int_{\Gamma_S} \phi \, ds \ (\forall \phi \in C_0^\infty(\Gamma_S) \text{ with } \phi \ge 0) \Big\},$$

and define (see [39])

$$\mathrm{Sup}_{\Gamma_S} \mu := \begin{cases} \inf M(\mu) & \text{if } M(\mu) \neq \emptyset, \\ +\infty & \text{if } M(\mu) = \emptyset. \end{cases}$$

Throughout this paper, $\boldsymbol{\tau} \cdot \boldsymbol{n} \ge 0$ (resp. $p \ge 0$) stands for $\mathrm{Sup}_{\Gamma_S}(-\boldsymbol{\tau} \cdot \boldsymbol{n}) \le 0$ (resp. $\mathrm{Sup}_{\Gamma_S}(-p) \le 0$) for any $\boldsymbol{\tau} \in V$ (resp. $p \in \Lambda_{-1/2}$).

2.1. The mixed variational inequality

Proposition 2.1. *Let $(\boldsymbol{\sigma}, u) \in \tilde{V} \times Q$ be the solution of* **(MVI)**. *If $u \in H^1(\Omega)$ and $\boldsymbol{\sigma} \cdot \boldsymbol{n} \in \Lambda$, then $(\boldsymbol{\sigma}, u)$ satisfies (3) where the Signorini condition (3d) holds a.e. on Γ_S.*

Proof. Taking $\boldsymbol{\tau} = \boldsymbol{\sigma} \pm \boldsymbol{\xi}$ ($\forall \boldsymbol{\xi} \in V_{\Gamma_S}$) into (4a) yields

$$(\boldsymbol{\sigma}, \boldsymbol{\xi}) + (u, \nabla \cdot \boldsymbol{\xi}) = 0 \qquad \forall \boldsymbol{\xi} \in V_{\Gamma_S}.$$

Under the assumption $u \in H^1(\Omega)$, by using the integration by parts, we see that

$$(\boldsymbol{\sigma}, \boldsymbol{\xi}) - (\nabla u, \boldsymbol{\xi}) + \langle \boldsymbol{\xi} \cdot \boldsymbol{n}, u \rangle_{\Gamma_D} = 0 \qquad \forall \boldsymbol{\xi} \in V_{\Gamma_S}. \tag{9}$$

Taking $\boldsymbol{\xi} \in V_{\Gamma_S} \subset V$ s.t. $\boldsymbol{\xi} \cdot \boldsymbol{n} = 0$, i.e. on Γ_D, we obtain (3a) in the weak sense. It follows from (3a) and (9) that

$$\langle \boldsymbol{\xi} \cdot \boldsymbol{n}, u \rangle_{\Gamma_D} = 0 \qquad \forall \boldsymbol{\xi} \in V_{\Gamma_S},$$

which implies $u = 0$ a.e. on Γ_D. Now applying the integration by parts to (4a) again, we get

$$(\boldsymbol{\sigma} - \nabla u, \boldsymbol{\tau} - \boldsymbol{\sigma}) + \langle (\boldsymbol{\tau} - \boldsymbol{\sigma}) \cdot \boldsymbol{n}, u \rangle_{\Gamma_S} \ge 0 \qquad \forall \boldsymbol{\tau} \in \tilde{V},$$

which implies (recalling that we have assumed $\boldsymbol{\sigma} \cdot \boldsymbol{n} \in L^2(\Gamma_S)$),

$$(\boldsymbol{\tau} \cdot \boldsymbol{n} - \boldsymbol{\sigma} \cdot \boldsymbol{n}, u)_{\Gamma_S} \ge 0 \qquad \forall \boldsymbol{\tau} \in \tilde{V} \cap \hat{V}. \tag{10}$$

Note that $(a, b)_{\Gamma_S} := \int_{\Gamma_S} ab \ \ d\Gamma$ ($\forall a, b \in L^2(\Gamma_S)$). Substituting $\boldsymbol{\tau} = 0, 2\boldsymbol{\sigma}$ into (10), we obtain

$$(\boldsymbol{\sigma} \cdot \boldsymbol{n}, u)_{\Gamma_S} = 0, \quad (\boldsymbol{\tau} \cdot \boldsymbol{n}, u)_{\Gamma_S} \ge 0 \quad \forall \boldsymbol{\tau} \in \tilde{V} \cap \hat{V},$$

i.e, $u \ge 0$ a.e on Γ_S and $\boldsymbol{\sigma} \cdot \boldsymbol{n} u = 0$ on Γ_S. Together with $\boldsymbol{\sigma} \cdot \boldsymbol{n} \ge 0$ on Γ_S and (4b), we conclude that $(\boldsymbol{\sigma}, u)$ satisfies (3). □

Remark 2.3. Proposition 2.1 says that $(\boldsymbol{\sigma}, u) \in \tilde{V} \times Q$ of (4) is a weak solution of (3). The regularity requirement of u of **(MVI)** is weaker compared with **(VI)**. It is known that **(VI)** admits a unique solution $u \in \tilde{X}$ (cf. [15]). If u is $H^2(\Omega)$-smooth, then u satisfies (1) with the Signorini condition (1c) holding a.e. on Γ_S, and $(\boldsymbol{\sigma} = \nabla u, u)$ satisfies (3) (also (4)).

Lemma 2.1. *There exists a constant $C_1 > 0$ such that, for any $f \in Q$ and $\xi \in \Lambda_{-1/2}$, one can find a $\boldsymbol{\sigma} \in V$ satisfying*

$$\nabla \cdot \boldsymbol{\sigma} = f \text{ in } \Omega, \quad \boldsymbol{\sigma} \cdot \boldsymbol{n} = \xi \text{ on } \Gamma_S, \tag{11a}$$

$$\|\boldsymbol{\sigma}\|_V \le C_1(\|f\|_Q + \|\xi\|_{\Lambda_{-1/2}}). \tag{11b}$$

Proof. Given $f \in Q$ and $\xi \in \Lambda_{-1/2}$, the elliptic problem

$$\Delta\phi = f \quad \text{in } \Omega, \quad \frac{\partial \phi}{\partial n} = \xi \quad \text{on } \Gamma_S, \quad \phi = 0 \quad \text{on } \Gamma_D, \tag{12}$$

admits a unique solution $\phi \in H^1_{\Gamma_D}(\Omega)$ with the boundedness

$$\|\phi\|_{H^1} \le C(\|f\|_Q + \|\xi\|_{\Lambda_{-1/2}}). \tag{13}$$

Taking $\boldsymbol{\sigma} = \nabla\phi$, and in view of (12) and (13), we conclude (11). □

Theorem 2.1 (*The Well-Posedness of* **(MVI)**). *Given $f \in Q$,* **(MVI)** *admits a unique solution $(\boldsymbol{\sigma}, u) \in \tilde{V} \times Q$ with the estimate*

$$\|\boldsymbol{\sigma}\|_V + \|u\|_Q \le C\|f\|_Q. \tag{14}$$

The unique existence of **(MVI)** has been proved by [32]. We would like to present an alternative proof revealing that $\boldsymbol{\sigma}$ can be seen as a projection onto a closed convex subspace of V.

Proof. (*Existence*) According to Lemma 2.1 (in the case of $\xi = 0$), for any $f \in Q$, there exists a $\boldsymbol{\sigma}_f \in V$ satisfying

$$\nabla \cdot \boldsymbol{\sigma}_v = v \text{ in } \Omega, \quad \boldsymbol{\sigma}_v \cdot \boldsymbol{n} = 0 \text{ on } \Gamma_S, \tag{15a}$$

$$\|\boldsymbol{\sigma}_v\|_V \le C_1 \|v\|_Q, \tag{15b}$$

where the constant C_1 is the same to that of (11b). Consider the following variational inequality:
For $f \in Q$, we define $\boldsymbol{\sigma}_f$ by (15). Find $\mathring{\boldsymbol{\sigma}} \in \mathring{\tilde{V}}$ such that

$$(\boldsymbol{\sigma}_f - \mathring{\boldsymbol{\sigma}}, \mathring{\boldsymbol{\tau}} - \mathring{\boldsymbol{\sigma}}) = (\boldsymbol{\sigma}_f - \mathring{\boldsymbol{\sigma}}, \mathring{\boldsymbol{\tau}} - \mathring{\boldsymbol{\sigma}})_V \le 0 \quad \forall \mathring{\boldsymbol{\tau}} \in \mathring{\tilde{V}},$$

where $(\boldsymbol{\sigma}, \boldsymbol{\tau})_V := (\boldsymbol{\sigma}, \boldsymbol{\tau}) + (\nabla \cdot \boldsymbol{\sigma}, \nabla \cdot \boldsymbol{\tau})$, for all $\boldsymbol{\sigma}, \boldsymbol{\tau} \in V$. Since $\mathring{\tilde{V}}$ is a closed convex subset of V, $\mathring{\boldsymbol{\sigma}}$ is the unique projection of $\boldsymbol{\sigma}_f$ onto $\mathring{\tilde{V}}$. Additionally, we have the boundedness

$$\|\mathring{\boldsymbol{\sigma}}\|_V \le \|\boldsymbol{\sigma}_f\|_V. \tag{16}$$

Set $\tilde{V}_f := \{\boldsymbol{\tau} \in V : \nabla \cdot \boldsymbol{\tau} = -f, \boldsymbol{\tau} \cdot \boldsymbol{n} \ge 0 \text{ on } \Gamma_S\}$ and $\boldsymbol{\sigma} := \mathring{\boldsymbol{\sigma}} - \boldsymbol{\sigma}_f \in \tilde{V}_f$. We find that

$$(\boldsymbol{\sigma}, \boldsymbol{\tau}_f - \boldsymbol{\sigma}) = (\boldsymbol{\sigma}, \boldsymbol{\tau}_f - \boldsymbol{\sigma})_V \ge 0 \quad \forall \boldsymbol{\tau}_f \in \tilde{V}_f, \tag{17}$$

which says that $\boldsymbol{\sigma}$ is the projection of $\mathbf{0}$ onto the closed convex subset $\tilde{V}_f$. It follows from (15) and $V_{\Gamma_S} \subset \hat{V} \subset V$ that

$$C_1^{-1} \|v\|_Q \le \sup_{\boldsymbol{\tau} \in V_{\Gamma_S}} \frac{(v, \nabla \cdot \boldsymbol{\tau})}{\|\boldsymbol{\tau}\|_V} \le \sup_{\boldsymbol{\tau} \in \hat{V}} \frac{(v, \nabla \cdot \boldsymbol{\tau})}{\|\boldsymbol{\tau}\|_V} \le \sup_{\boldsymbol{\tau} \in V} \frac{(v, \nabla \cdot \boldsymbol{\tau})}{\|\boldsymbol{\tau}\|_V}. \tag{18}$$

The above inf-sup condition ensures the unique existence of $u \in Q$ satisfying

$$(u, \nabla \cdot \boldsymbol{\xi}) = (-\boldsymbol{\sigma}, \boldsymbol{\xi}) \quad \forall \boldsymbol{\xi} \in V_{\Gamma_S}, \tag{19a}$$

$$\|u\|_Q \le C_1 \sup_{\boldsymbol{\xi} \in V_{\Gamma_S}} \frac{(u, \nabla \cdot \boldsymbol{\xi})}{\|\boldsymbol{\xi}\|_V} = C_1 \sup_{\boldsymbol{\xi} \in V_{\Gamma_S}} \frac{(-\boldsymbol{\sigma}, \boldsymbol{\xi})}{\|\boldsymbol{\xi}\|_V} \le C_1 \|\boldsymbol{\sigma}\|_Q. \tag{19b}$$

It remains to verify that the above $(\boldsymbol{\sigma}, u)$ satisfies **(MVI)**. By using Lemma 2.1, one can show that every $\boldsymbol{\tau} \in \tilde{V}$ has a decomposition $\boldsymbol{\tau} = \boldsymbol{\tau}_f + \boldsymbol{\xi}$ with $\boldsymbol{\tau}_f \in \tilde{V}_f$ and $\boldsymbol{\xi} \in V_{\Gamma_S}$. Then, it follows from (17) and (19a) that

$$\begin{aligned} 0 \le (\boldsymbol{\sigma}, \boldsymbol{\tau}_f - \boldsymbol{\sigma}) &= (\boldsymbol{\sigma}, \boldsymbol{\tau} - \boldsymbol{\sigma}) - (\boldsymbol{\sigma}, \boldsymbol{\xi}) = (\boldsymbol{\sigma}, \boldsymbol{\tau} - \boldsymbol{\sigma}) + (u, \nabla \cdot \boldsymbol{\xi}) \\ &= (\boldsymbol{\sigma}, \boldsymbol{\tau} - \boldsymbol{\sigma}) + (u, \nabla \cdot (\boldsymbol{\tau} - \boldsymbol{\tau}_f)) \\ &= (\boldsymbol{\sigma}, \boldsymbol{\tau} - \boldsymbol{\sigma}) + (u, \nabla \cdot (\boldsymbol{\tau} - \boldsymbol{\sigma})) \quad (\text{by } \nabla \cdot \boldsymbol{\sigma} = -f = \nabla \cdot \boldsymbol{\tau}_f). \end{aligned}$$

Therefore, it has been demonstrated that $(\boldsymbol{\sigma}, u)$ is a solution of **(MVI)**. A combination of (15), (16) and (19b) results the *a-prior* estimate (14).

(*Uniqueness*) Suppose **(MVI)** admits two solutions $(\boldsymbol{\sigma}_1, u_1)$ and $(\boldsymbol{\sigma}_2, u_2)$. We have $\nabla \cdot \boldsymbol{\sigma}_1 = \nabla \cdot \boldsymbol{\sigma}_2 = f$ and

$$(\boldsymbol{\sigma}_1, \boldsymbol{\tau} - \boldsymbol{\sigma}_1) + (u_1, \nabla \cdot (\boldsymbol{\tau} - \boldsymbol{\sigma}_1)) \ge 0 \quad \forall \boldsymbol{\tau} \in \tilde{V}, \tag{20a}$$

$$(\boldsymbol{\sigma}_2, \boldsymbol{\tau} - \boldsymbol{\sigma}_2) + (u_2, \nabla \cdot (\boldsymbol{\tau} - \boldsymbol{\sigma}_2)) \ge 0 \quad \forall \boldsymbol{\tau} \in \tilde{V}. \tag{20b}$$

Substituting $\boldsymbol{\tau} = \boldsymbol{\sigma}_2$ into (20a) and $\boldsymbol{\tau} = \boldsymbol{\sigma}_1$ into (20b), and summing up two inequalities, we obtain

$$\|\boldsymbol{\sigma}_1 - \boldsymbol{\sigma}_2\|_Q^2 = 0.$$

Thus, $\boldsymbol{\sigma}_1 = \boldsymbol{\sigma}_2$. The inf-sup condition (18) then implies $u_1 = u_2$. □

2.2. The mixed Lagrange multiplier formulations

Proposition 2.2. **(MVI)**, **(MLM-I)** *and* **(MLM-II)** *are equivalent.*

Proof. The proof is divided into four steps. The equivalence between **(MVI)** and **(MLM-I)** is demonstrated in (Step 1) and (Step 2). (Step 3) and (Step 4) show that **(MLM-I)** equals **(MLM-II)**.

(Step 1) Let $(\boldsymbol{\sigma}, u, p) \in \tilde{V} \times Q \times \Lambda_{1/2}$ be the solution of **(MLM-I)**. It is easy to see that (5a) and (5c) yields (4a), i.e., $(\boldsymbol{\sigma}, u)$ solves **(MVI)**.

(Step 2) Let $(\boldsymbol{\sigma}, u) \in \tilde{V} \times Q$ be the solution of **(MVI)**. According to Lemma 2.1 (in the case of $f = 0$), for any $\mu \in \Lambda_{-1/2}$, there exist a $\boldsymbol{\tau}_\mu \in \mathring{V}$ such that

$$\boldsymbol{\tau}_\mu \cdot \boldsymbol{n} = \mu \text{ on } \Gamma_S, \tag{21a}$$

$$\|\boldsymbol{\tau}_\mu\|_V \le C_1 \|\mu\|_{\Lambda_{-1/2}}. \tag{21b}$$

Following from (21) and the duality of $\Lambda_{1/2}$ and $\Lambda_{-1/2}$, we have the inf-sup condition

$$\begin{aligned}\|p\|_{\Lambda_{1/2}} &= \sup_{\mu\in\Lambda_{-1/2}} \frac{\langle \mu, p\rangle_{\Gamma_S}}{\|\mu\|_{\Lambda_{-1/2}}} \le \sup_{\tau\in V} \frac{\langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S}}{\|\boldsymbol{\tau}\cdot\boldsymbol{n}\|_{\Lambda_{-1/2}}} \\ &\le C_1 \sup_{\tau\in\tilde{V}} \frac{\langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S}}{\|\boldsymbol{\tau}\|_V} \le C_1 \sup_{\tau\in V} \frac{\langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S}}{\|\boldsymbol{\tau}\|_V} \quad (\forall p\in\Lambda_{1/2}).\end{aligned} \tag{22}$$

By using (22), there exists a unique $p\in\Lambda_{1/2}$ satisfying

$$\langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S} = (\boldsymbol{\sigma},\boldsymbol{\tau}) + (u, \nabla\cdot\boldsymbol{\tau}) \quad \forall \boldsymbol{\tau}\in V,$$

with the boundedness

$$C_1^{-1}\|p\|_{\Lambda_{1/2}} \le \sup_{\tau\in V} \frac{(\boldsymbol{\sigma},\boldsymbol{\tau}) + (u,\nabla\cdot\boldsymbol{\tau})}{\|\boldsymbol{\tau}\|_V} \le \|\boldsymbol{\sigma}\|_Q + \|u\|_Q.$$

We have shown that (5a) holds. And (5c) is a direct result from (4a) and (5a). Hence, $(\boldsymbol{\sigma}, u, p)$ is a solution of **(MLM-I)**.

(Step 3) Let $(\boldsymbol{\sigma}, u, p)\in V\times Q\times\tilde{\Lambda}_{1/2}$ be the solution of **(MLM-II)**. Substituting $q = 0, 2p$ into (6c) yields

$$\langle \boldsymbol{\sigma}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S} = 0, \tag{23a}$$

$$\langle \boldsymbol{\sigma}\cdot\boldsymbol{n}, q\rangle_{\Gamma_S} \ge 0 \quad \forall q\in\tilde{\Lambda}_{1/2}, \tag{23b}$$

Note that (23b) implies $\boldsymbol{\sigma}\in\tilde{V}$. Since $C_0^\infty(\Gamma_S)$ is dense in $\Lambda_{1/2}$, there exist a sequence $p_n\in C_0^\infty(\Gamma_S)$ with $p_n\ge 0$ and $p_n\to p$ in $\Lambda_{1/2}$. For any $\boldsymbol{\tau}\in\tilde{V}$, in view of

$$\langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p_n\rangle_{\Gamma_S} \ge 0, \quad \langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p_n\rangle_{\Gamma_S} \to \langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S},$$

we have

$$\langle \boldsymbol{\tau}\cdot\boldsymbol{n}, p\rangle_{\Gamma_S} \ge 0 \quad \forall \boldsymbol{\tau}\in\tilde{V}. \tag{24}$$

Combining (23a) and (24), we obtain (5c). Hence, $(\boldsymbol{\sigma}, u, p)\in\tilde{V}\times Q\times\Lambda_{1/2}$ solves **(MLM-I)**.

(Step 4) Let $(\boldsymbol{\sigma}, u, p)\in\tilde{V}\times Q\times\Lambda_{1/2}$ be the solution of **(MLM-I)**. Substituting $\boldsymbol{\tau} = 0, 2\boldsymbol{\sigma}$ into (5c) results (23a) and (24). Note that (24) implies $p\ge 0$ a.e on Γ_S, i.e. $p\in\tilde{\Lambda}_{1/2}$. In order to show that $(\boldsymbol{\sigma}, u, p)$ satisfies (6), it suffices to verify (23b). For any $q\in\tilde{\Lambda}_{1/2}$, there is a sequence $q_n\in C_0^\infty(\Gamma_S)$ with $q_n\ge 0$ and $q_n\to q$ in $\Lambda_{1/2}$. Since $\boldsymbol{\sigma}\cdot\boldsymbol{n}\ge 0$ on Γ_S, we have

$$0 \le \langle \boldsymbol{\sigma}\cdot\boldsymbol{n}, q_n\rangle_{\Gamma_S} \to \langle \boldsymbol{\sigma}\cdot\boldsymbol{n}, q\rangle_{\Gamma_S},$$

which yields (23b). Hence, $(\boldsymbol{\sigma}, u, p)$ solves **(MLM-II)**. □

Remark 2.4. Under the assumption that $\boldsymbol{\sigma}\cdot\boldsymbol{n}\in\Lambda$, we can replace the dual-product $\langle\cdot,\cdot\rangle_{\Gamma_S}$ of (5c) with the inner-product $(\cdot,\cdot)_{\Gamma_S}$. Therefore, we have: for any $\boldsymbol{\tau}\in\tilde{V}$ with $\boldsymbol{\tau}\cdot\boldsymbol{n}\in\Lambda$,

$$(\boldsymbol{\tau}\cdot\boldsymbol{n} - \boldsymbol{\sigma}\cdot\boldsymbol{n}, -\rho p + \boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} \le 0 \quad (\forall\rho\ge 0),$$

which is equivalent to

$$\boldsymbol{\sigma}\cdot\boldsymbol{n} = P_{\tilde{\Lambda}}(-\rho p + \boldsymbol{\sigma}\cdot\boldsymbol{n}) = \max(0, -\rho p + \boldsymbol{\sigma}\cdot\boldsymbol{n}). \tag{25}$$

Analogously, since $\tilde{\Lambda}_{1/2}$ is dense in $\tilde{\Lambda}$, if $\boldsymbol{\sigma}\cdot\boldsymbol{n}\in\Lambda$, then we can write (6c) into

$$(-\rho\boldsymbol{\sigma}\cdot\boldsymbol{n} + p - p, q - p)_{\Gamma_S} \le 0 \quad (\forall q\in\tilde{\Lambda},\ \forall\rho\ge 0),$$

or equivalently,

$$p = P_{\tilde{\Lambda}}(-\rho\boldsymbol{\sigma}\cdot\boldsymbol{n} + p) = \max(0, -\rho\boldsymbol{\sigma}\cdot\boldsymbol{n} + p). \tag{26}$$

These two projection formulations (25) and (26) are the key to designing the Active/Inactive set algorithms presented in Section 3.

Remark 2.5. In case we possess solely the regularity $\boldsymbol{\sigma}\cdot\boldsymbol{n}\in\Lambda_{-1/2}$, the projection formulas (25) and (26) will require modification in the following way. According to the Riesz representation theorem, for any $\xi\in\Lambda_{-1/2} = \Lambda_{1/2}^*$, there exists a unique $r\in\Lambda_{1/2}$ such that

$$\langle \xi, q\rangle_{\Gamma_S} = (r, q)_{\Lambda_{1/2}} \quad (\forall q\in\Lambda_{1/2}), \tag{27}$$

where $(\cdot,\cdot)_{\Lambda_{1/2}}$ represents the inner-product of the Hilbert space $\Lambda_{1/2}$. Now, we define the operator

$$R : \Lambda_{-1/2}\to\Lambda_{1/2}, \quad R\xi = r,$$

where r satisfies (27) for all $\xi\in\Lambda_{-1/2}$. In fact, R is the duality mapping, and it is well-known that R is linear and one-to-one, and satisfies $\|R\|_{\mathcal{L}(\Lambda_{-1/2},\Lambda_{1/2})} = 1$. As a result, we can express (5c) by using R and $(\cdot,\cdot)_{\Lambda_{1/2}}$, that is

$$(R\boldsymbol{\tau}\cdot\boldsymbol{n} - R\boldsymbol{\sigma}\cdot\boldsymbol{n}, -\rho p + R\boldsymbol{\sigma}\cdot\boldsymbol{n} - R\boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Lambda_{1/2}} \le 0 \quad (\forall\boldsymbol{\tau}\in\tilde{V},\ \forall\rho\ge 0),$$

or equivalently,

$$R\boldsymbol{\sigma}\cdot\boldsymbol{n} = P_{\tilde{\Lambda}_{1/2}}(-\rho p + R\boldsymbol{\sigma}\cdot\boldsymbol{n}) \quad (\forall \rho \geq 0), \tag{28}$$

where $P_{\tilde{\Lambda}_{1/2}}$ is the projection operator from $\Lambda_{1/2}$ onto the convex subspace $\tilde{\Lambda}_{1/2}$ under the inner product $(\cdot,\cdot)_{\Lambda_{1/2}}$.

Similarly, from (6c), we have

$$(-\rho R\boldsymbol{\sigma}\cdot\boldsymbol{n} + p - p, q - p)_{\Lambda_{1/2}} \leq 0 \quad (\forall q \in \tilde{\Lambda}_{1/2},\ \forall \rho \geq 0),$$

or equivalently,

$$p = P_{\tilde{\Lambda}_{1/2}}(-\rho R\boldsymbol{\sigma}\cdot\boldsymbol{n} + p). \tag{29}$$

In fact, the modified Uzawa algorithm proposed by [32] is based on a discrete version of (29) utilizing the boundary element method. In the present work, we assume that $\boldsymbol{\sigma}\cdot\boldsymbol{n} \in \Lambda$ and use the projection (25) and (26) to design the iteration scheme in the finite element discretization.

3. The finite element approximation

In this section, we consider the finite element approximation for **(MVI)** by using the RT0/P0-element. To avoid the issue of geometric domain approximation, we assume that Ω is a polygon ($d = 2$) or polyhedron ($d = 3$) and introduce a quasi-uniform regular triangulation $\mathcal{T}_h$ to Ω with mesh size $h = \max_{K\in\mathcal{T}_h} \mathrm{diam}(T)$. We denote by $\mathcal{E}_h$ the set of elements faces, by $\mathcal{E}_{h,\Gamma_S}$ the mesh on Γ_S inherited from $\mathcal{T}_h$.

3.1. The discrete *MVI*

We introduce the finite element spaces:

$$\begin{aligned}
&V_h := \{\boldsymbol{\tau}_h \in V : \boldsymbol{\tau}_h|_T \in RT_0(T) \quad (\forall T \in \mathcal{T}_h)\}, \quad \tilde{V}_h := \{\boldsymbol{\tau}_h \in V_h : \boldsymbol{\tau}_h\cdot\boldsymbol{n} \geq 0 \text{ on } \Gamma_S\},\\
&V_{h,\Gamma_S} := \{\boldsymbol{\tau}_h \in V_h : \boldsymbol{\tau}_h\cdot\boldsymbol{n} = 0 \text{ on } \Gamma_S\}, \quad \mathring{V}_h := \{\boldsymbol{\tau}_h \in V_h : \nabla\cdot\boldsymbol{\tau}_h|_T = 0 \quad (\forall T \in \mathcal{T}_h)\},\\
&\Lambda_h := \{p_h \in \Lambda : p_h|_e \in P_0(e) \quad (\forall e \in \mathcal{E}_{h,\Gamma_S})\} = \{\boldsymbol{\tau}_h\cdot\boldsymbol{n}|_{\Gamma_S} : \boldsymbol{\tau}_h \in V_h\},\\
&\tilde{\Lambda}_h := \{p_h \in \Lambda : p_h \geq 0\}, \quad Q_h := \{v_h \in Q : v_h|_T \in P_0(T) \quad (\forall T \in \mathcal{T}_h)\},
\end{aligned}$$

where $P_0(T)$ and $P_0(e)$ denote the set of piecewise constant functions in T and e, respectively, and

$$\begin{aligned}
&RT_0(T) := \{(a,b)^\top + c(x,y)^\top : a,b,c \in \mathbb{R},\ (x,y)^\top \in T\} \quad (d = 2),\\
&RT_0(T) := \{(a,b,c)^\top + d(x,y,z)^\top : a,b,c,d \in \mathbb{R},\ (x,y,z)^\top \in T\} \quad (d = 3).
\end{aligned}$$

Now, let us state the discrete mixed variational inequality:

(MVI)$_h$ Find $(\boldsymbol{\sigma}_h, u_h) \in \tilde{V}_h \times Q_h$ such that

$$(\boldsymbol{\sigma}_h, \boldsymbol{\tau}_h - \boldsymbol{\sigma}_h) + (u_h, \nabla\cdot(\boldsymbol{\tau}_h - \boldsymbol{\sigma}_h)) \geq 0 \qquad \forall \boldsymbol{\tau}_h \in \tilde{V}_h, \tag{30a}$$

$$(\nabla\cdot\boldsymbol{\sigma}_h, v_h) + (f, v_h) = 0 \qquad \forall v_h \in Q_h. \tag{30b}$$

Remark 3.1. Note that $\mathrm{div}\, V_h = Q_h$, and (30b) is equivalent to $-\nabla\cdot\boldsymbol{\sigma}_h = f_h := P_h f$, where $P_h : Q \to Q_h$ is the L^2-projection defined by

$$(f - P_h f, v_h) = 0 \quad (\forall v_h \in Q_h). \tag{31}$$

The following estimates hold for P_h:

$$\|P_h f\|_Q \leq \|f\|_Q \quad (\forall f \in Q), \qquad \|P_h f - f\|_Q \leq Ch\|f\|_{H^1(\Omega)} \quad (\forall f \in H^1(\Omega)).$$

Set $W(T) := \{\boldsymbol{\tau} \in L^s(T)^d : \nabla\cdot\boldsymbol{\tau} \in L^2(T)\}$ for some $s \in (2,\infty]$.

Remark 3.2. For any $q_h \in L^2(T)$, the integral $\int_e \boldsymbol{\tau}\cdot\boldsymbol{n} q_h\, ds$ cannot be calculated in general if $\boldsymbol{\tau} \in H(\mathrm{div}, T)$ which means $\boldsymbol{\tau}\cdot\boldsymbol{n}$ is defined only in $H^{-\frac{1}{2}}(\partial T)$ (cf. [30, p103]). To define the interpolation operator ρ_h, $\boldsymbol{\tau}$ needs to be smoother than $H(\mathrm{div}, T)$.

Now we introduce the interpolation operator:

$$\rho_h : W(T) \to RT_0(T), \qquad \int_e (\boldsymbol{\tau} - \rho_h\boldsymbol{\tau})\cdot\boldsymbol{n}\, ds = 0 \quad (\forall e \in \partial T). \tag{32}$$

We set $W := V \cap L^s(\Omega)^d$ and define the interpolation operator:

$$\Pi_h : W \to V_h, \qquad \Pi_h\boldsymbol{\tau}|_T = \rho_h(\boldsymbol{\tau}|_T) \quad (\forall T \in \mathcal{T}_h). \tag{33}$$

For ρ_h and Π_h, the following properties hold:

$$\operatorname{div}(\rho_h\boldsymbol{\tau}) = P_{h,T}(\operatorname{div}\boldsymbol{\tau}) \quad (\forall \boldsymbol{\tau} \in W(T)), \tag{34a}$$

$$\operatorname{div}(\Pi_h\boldsymbol{\tau}) = P_h(\operatorname{div}\boldsymbol{\tau}) \quad (\forall \boldsymbol{\tau} \in W), \tag{34b}$$

$$\|\boldsymbol{\tau} - \rho_h\boldsymbol{\tau}\|_{L^2(T)} \le Ch\|\boldsymbol{\tau}\|_{H^1(T)}, \quad \|\nabla\cdot(\boldsymbol{\tau} - \rho_h\boldsymbol{\tau})\|_{L^2(T)} \le Ch\|\nabla\cdot\boldsymbol{\tau}\|_{H^1(T)}, \tag{34c}$$

$$\|\boldsymbol{\tau} - \Pi_h\boldsymbol{\tau}\|_Q \le Ch\|\boldsymbol{\tau}\|_{H^1(\Omega)}, \quad \|\nabla\cdot(\boldsymbol{\tau} - \Pi_h\boldsymbol{\tau})\|_Q \le Ch\|\nabla\cdot\boldsymbol{\tau}\|_{H^1(\Omega)}, \tag{34d}$$

where $P_{h,T}$ represents the L^2-projection from $L^2(T)$ to $P_0(T)$.

Remark 3.3. The properties (34a) and (34b) can also be expressed as

$$\begin{array}{ccc} W(T) & \xrightarrow{\operatorname{div}} & L^2(T) \\ {\scriptstyle\rho_h}\downarrow & & \downarrow{\scriptstyle P_{h,T}} \\ RT_0(T) & \xrightarrow{\operatorname{div}} & P_0(T) \end{array} \qquad \begin{array}{ccc} W & \xrightarrow{\operatorname{div}} & Q \\ {\scriptstyle\Pi_h}\downarrow & & \downarrow{\scriptstyle P_h} \\ V_h & \xrightarrow{\operatorname{div}} & Q_h \end{array}$$

Furthermore, we define the L^2-projection operator $P_{\Lambda_h} : \Lambda \to \Lambda_h$ by

$$(P_{\Lambda_h}p - p, q_h)_{\Gamma_S} = 0 \quad (\forall q_h \in \Lambda_h), \tag{35}$$

which satisfies

$$\|P_{\Lambda_h}p\|_\Lambda \le \|p\|_\Lambda \qquad (\forall p \in \Lambda),$$

$$\|P_{\Lambda_h}p - p\|_\Lambda \le Ch\|p\|_{H^1(\Gamma_S)} \qquad (\forall p \in H^1(\Gamma_S)).$$

The discrete mixed Lagrange multiplier formulations for **(MLM-I)** and **(MLM-II)** are presented as follows.

(MLM-I)$_h$ Find $(\boldsymbol{\sigma}_h, u_h, p_h) \in \tilde{V}_h \times Q_h \times \Lambda_h$ such that

$$(\boldsymbol{\sigma}_h, \boldsymbol{\tau}_h) + (u_h, \nabla\cdot\boldsymbol{\tau}_h) - (p_h, \boldsymbol{\tau}_h\cdot\boldsymbol{n})_{\Gamma_S} = 0 \qquad \forall \boldsymbol{\tau}_h \in V_h, \tag{36a}$$

$$(\nabla\cdot\boldsymbol{\sigma}_h, v_h) + (f, v_h) = 0 \qquad \forall v_h \in Q_h, \tag{36b}$$

$$(\boldsymbol{\tau}_h\cdot\boldsymbol{n} - \boldsymbol{\sigma}_h\cdot\boldsymbol{n}, p_h)_{\Gamma_S} \ge 0 \qquad \forall \boldsymbol{\tau}_h \in \tilde{V}_h. \tag{36c}$$

(MLM-II)$_h$ Find $(\boldsymbol{\sigma}_h, u_h, p_h) \in V_h \times Q_h \times \tilde{\Lambda}_h$ such that

$$(\boldsymbol{\sigma}_h, \boldsymbol{\tau}_h) + (u_h, \nabla\cdot\boldsymbol{\tau}_h) - (p_h, \boldsymbol{\tau}_h\cdot\boldsymbol{n})_{\Gamma_S} = 0 \qquad \forall \boldsymbol{\tau}_h \in V_h, \tag{37a}$$

$$(\nabla\cdot\boldsymbol{\sigma}_h, v_h) + (f, v_h) = 0 \qquad \forall v_h \in Q_h, \tag{37b}$$

$$(q_h - p_h, \boldsymbol{\sigma}_h\cdot\boldsymbol{n})_{\Gamma_S} \ge 0 \qquad \forall q_h \in \tilde{\Lambda}_h. \tag{37c}$$

Proposition 3.1. **(MVI)**$_h$, **(MLM-I)**$_h$ *and* **(MLM-II)**$_h$ *are equivalent.*

Proof. The proof is divided into four steps. In (Step 1) and (Step 2), we show that **(MLM-I)**$_h$ and **(MVI)**$_h$ are equivalent. (Step 3) and (Step 4) illustrate that **(MLM-I)**$_h$ equals **(MLM-II)**$_h$.

(Step 1) It is apparent that the solution of **(MLM-I)**$_h$ also satisfies **(MVI)**$_h$.

(Step 2) Let $(\boldsymbol{\sigma}_h, u_h)$ be the solution of **(MVI)**$_h$. Given $p_h \in \Lambda_h \subset \Lambda$, there exists a $w \in W^{1,s}(\Omega)$ for some $s \in (2, \infty)$ satisfying

$$\Delta w = 0 \text{ in } \Omega, \quad w = 0 \text{ on } \Gamma_D, \quad \nabla w\cdot\boldsymbol{n} = p_h \text{ on } \Gamma_S, \tag{38a}$$

$$\|w\|_{W^{1,s}(\Omega)} \le C_2\|p_h\|_\Lambda. \tag{38b}$$

In view of $\nabla\cdot\boldsymbol{\tau} = 0$ and $\nabla w \in L^s(\Omega)$ (see Remark 3.2 we need $s > 2$), we set $\boldsymbol{\tau} := \nabla w \in W$ and $\boldsymbol{\tau}_h := \Pi_h\boldsymbol{\tau} \in V_h$, which satisfies

$$\boldsymbol{\tau}_h\cdot\boldsymbol{n} = \boldsymbol{\tau}\cdot\boldsymbol{n} = p_h \text{ on } e \in \mathcal{E}_{h,\Gamma_S}, \text{ where } \boldsymbol{n} \text{ is the unit normal vector to } e, \tag{39a}$$

$$\nabla\cdot\boldsymbol{\tau}_h = \nabla\cdot\boldsymbol{\tau} = 0, \tag{39b}$$

$$\|\boldsymbol{\tau}_h\|_V \le C\|\boldsymbol{\tau}\|_W = C\|\nabla w\|_{L^s(\Omega)} \le C_3\|p_h\|_\Lambda \quad (C_3 := CC_2). \tag{39c}$$

Following from (39), we have the inf-sup condition

$$\|p_h\|_\Lambda = \sup_{q_h\in\Lambda_h} \frac{(p_h, q_h)_{\Gamma_S}}{\|q_h\|_\Lambda} \le C_3 \sup_{\boldsymbol{\tau}_h\in\tilde{V}_h} \frac{(p_h, \boldsymbol{\tau}_h\cdot\boldsymbol{n})_{\Gamma_S}}{\|\boldsymbol{\tau}_h\|_V} \le C_3 \sup_{\boldsymbol{\tau}_h\in V_h} \frac{(p_h, \boldsymbol{\tau}_h\cdot\boldsymbol{n})_{\Gamma_S}}{\|\boldsymbol{\tau}_h\|_V} \quad (\forall p_h \in \Lambda_h), \tag{40}$$

which implies the unique existence of $p_h \in \Lambda_h$ such that (36a) holds. Moreover, (36c) is a direct result from (36a) and (30a). Therefore, $(\boldsymbol{\sigma}_h, u_h, p_h)$ solves (36).

(Step 3) We assert that the solution of **(MLM-I)**$_h$ also satisfies **(MLM-II)**$_h$. It suffices to show that $\boldsymbol{\sigma}_h \in \tilde{V}_h$ and (36c) implies $p_h \geq 0$ and (37c). Substituting $\boldsymbol{\tau}_h = 0, 2\boldsymbol{\sigma}_h$ into (36c) yeilds

$$(\boldsymbol{\sigma}_h \cdot \boldsymbol{n}, p_h)_{\Gamma_S} = 0, \tag{41a}$$

$$(\boldsymbol{\tau}_h \cdot \boldsymbol{n}, p_h)_{\Gamma_S} \geq 0 \quad (\forall \boldsymbol{\tau}_h \in \tilde{V}_h), \tag{41b}$$

where (41b) is a result from (36c) and (41a). Note that $\boldsymbol{\tau}_h|_T \in RT_0(T)$ implies $\boldsymbol{\tau}_h \cdot \boldsymbol{n}$ is a constant on each element face e (see [30, Proposition 2.3.3 or Example 2.3.1]). Since p_h is also piecewise constant on Γ_S and $\boldsymbol{\tau}_h \cdot \boldsymbol{n} \geq 0$ for any $\boldsymbol{\tau}_h \in \tilde{V}_h$, we conclude $p_h \geq 0$ from (41b). Together with (41a), we conclude (37c).

(Step 4) Conversely, the solution of **(MLM-II)**$_h$ is also a solution of **(MLM-I)**$_h$. We only need to show that $p_h \in \tilde{\Lambda}_h$ and (37c) implies $\boldsymbol{\sigma}_h \in \tilde{V}_h$ and (36c). Substituting $q_h = 0, 2p_h$ into (37c) gives (41a) and

$$(q_h, \boldsymbol{\sigma}_h \cdot \boldsymbol{n})_{\Gamma_S} \geq 0 \quad (\forall q_h \in \tilde{\Lambda}_h). \tag{42}$$

For RT0-element, we see that $\boldsymbol{\sigma}_h \cdot \boldsymbol{n}$ is piecewise constant on Γ_S. Combined with q_h is piecewise constant and $q_h > 0$ on Γ_S, (42) implies that $\boldsymbol{\sigma}_h \cdot \boldsymbol{n} \geq 0$ on Γ_S. Together with (41a), we get (36c). □

In the continuous sense, if the solution $(\boldsymbol{\sigma}, u, p)$ of **(MLM-I)** satisfies $u \in H^1(\Omega)$, then by using the integration by parts, one can verify that $u|_{\Gamma_S} = p$. In the discrete sense, we address in the following remark that $u_h|_{\Gamma_S} = p_h + O(\|\boldsymbol{\sigma}_h\|_{L^\infty})h$, where $(\boldsymbol{\sigma}_h, u_h, p_h)$ is the solution of **(MLM-I)**$_h$.

Remark 3.4. For any face $e \subset \Gamma_S$, we denote by T_e the triangle with the boundary element e, by $\{x_i\}_{i=1}^{d+1}$ the vertices of T_e, and by $\{e_i\}_{i=1}^{d+1}$ the faces of T_e, where the vertex x_i is opposite to the face e_i. Note that $\{\boldsymbol{\phi}_i(x) = h_{e_i}^{-1}(x - x_i)\}_{i=1}^{d+1}$ is a basis of $RT_0(T_e)$, where h_{e_i} is the length of the altitude from x_i to e_i. Without loss of generality, we assume that $e_1 = e \subset \Gamma_S$. So we have $\boldsymbol{\phi}_1(x) \cdot \boldsymbol{n}_e = 1$ and $\boldsymbol{\phi}_i(x) \cdot \boldsymbol{n}_e = 0$ for all $x \in e$ $(i = 2, 3, \ldots, d+1)$, where $\boldsymbol{n}_e$ denotes the unit outer normal to e. Noting that p_h and u_h are piecewise constant functions, we have

$$p_h(x) = p_e \quad (\forall x \in e), \quad u_h(x) = u_{T_e} \quad (\forall x \in T_e),$$
$$\boldsymbol{\sigma}_h = \sum_{i=1}^{d+1} \sigma_i \boldsymbol{\phi}_i(x) \quad (\forall x \in T_e),$$

where $p_e, u_{T_e}, \sigma_1, \sigma_2$, and σ_3 are five constants. Substituting $\boldsymbol{\tau} = \boldsymbol{\phi}_1(x)1_{T_e}$ into (36a), we get

$$\begin{aligned} 0 &= (\boldsymbol{\sigma}_h, \boldsymbol{\phi}_1(x))_{T_e} + \int_{Te} u_{T_e} \frac{d}{h_{e_1}}\, dx - \int_e p_e \cdot 1 \, d\Gamma \\ &= \sum_{i=1}^{d+1} \sigma_i (\boldsymbol{\phi}_i(x), \boldsymbol{\phi}_1(x))_{T_e} + \frac{d|T_e|}{h_e} u_{T_e} - |e| p_e \end{aligned}$$

which yields

$$p_e = \frac{d|T_e|}{h_e|e|} u_{T_e} + O(\max_{1 \leq i \leq d+1} |\sigma_i|)h_e = u_{T_e} + O(\|\sigma_i\|_{L^\infty(T_e)})h_e.$$

Next, we turn to demonstrate the unique existence of **(MVI)**$_h$. Given $f \in Q$, let w be the unique solution to the following elliptic problem:

$$\Delta w = f_h := P_h f \text{ in } \Omega, \quad w = 0 \text{ on } \Gamma_D, \quad \nabla w \cdot \boldsymbol{n} = 0 \text{ on } \Gamma_S.$$

We have $w \in W^{1,s}(\Omega)$ with $\|w\|_{W^{1,s}(\Omega)} \leq C\|f_h\|_Q$ (for some $s \in (2, \infty)$). Now we set

$$\boldsymbol{\sigma}_{h,f} := \Pi_h(\nabla w) \in V_{h,\Gamma_S},$$

which satisfies

$$\nabla \cdot \boldsymbol{\sigma}_{h,f} = f_h \quad (\text{i.e. } (\nabla \cdot \boldsymbol{\sigma}_{h,f}, v_h) = (f, v_h) = (f_h, v_h) \quad \forall v_h \in Q_h),$$
$$\|\boldsymbol{\sigma}_{h,f}\|_V = C\|f_h\|_Q \leq C\|f\|_Q.$$

Consider the following variational inequality:
(MVI)$_h^\circ$ Find $\mathring{\boldsymbol{\sigma}}_h \in \mathring{\tilde{V}}_h := \mathring{V}_h \cap \tilde{V}_h$ such that

$$(\mathring{\boldsymbol{\sigma}}_h, \mathring{\boldsymbol{\tau}}_h - \mathring{\boldsymbol{\sigma}}_h) \geq (\boldsymbol{\sigma}_{h,f}, \mathring{\boldsymbol{\tau}}_h - \mathring{\boldsymbol{\sigma}}_h) \quad (\forall \mathring{\boldsymbol{\tau}}_h \in \mathring{\tilde{V}}_h). \tag{43}$$

Remark 3.5. The following is inf-sup condition (the discrete version of (18))

$$\|u_h\|_Q \leq C_4 \sup_{\boldsymbol{\tau}_h \in V_{h,\Gamma_S}} \frac{(u_h, \nabla \cdot \boldsymbol{\tau}_h)}{\|\boldsymbol{\tau}_h\|_V} \leq C_4 \sup_{\boldsymbol{\tau}_h \in V_h} \frac{(u_h, \nabla \cdot \boldsymbol{\tau}_h)}{\|\boldsymbol{\tau}_h\|_V}. \tag{44}$$

The proof is similar to [30, Proposition 5.4.3], where $\boldsymbol{\tau}_h \in V_h$ instead of V_{h,Γ_S}. Obviously, from (34b) and (34d) we have

$$(\nabla\cdot\boldsymbol{\tau}-\nabla\cdot\Pi_h\boldsymbol{\tau},v_h)=0\quad\forall v_h\in Q_h,\quad \|\Pi_h\boldsymbol{\tau}\|_V\le\tilde{C}\|\boldsymbol{\tau}\|_V. \tag{45}$$

Given $v_h\in Q_h$, it follows from (18) and (45) that

$$\begin{aligned}\|v_h\|_Q &\le C_1\sup_{\boldsymbol{\tau}\in V_{\Gamma_S}}\frac{(v_h,\nabla\cdot\boldsymbol{\tau})}{\|\boldsymbol{\tau}\|_V}=C_1\sup_{\boldsymbol{\tau}\in V_{\Gamma_S}}\frac{(v_h,\nabla\cdot\Pi_h\boldsymbol{\tau})}{\|\boldsymbol{\tau}\|_V}\le C_1\tilde{C}\sup_{\boldsymbol{\tau}\in V_{\Gamma_S}}\frac{(v_h,\nabla\cdot\Pi_h\boldsymbol{\tau})}{\|\Pi_h\boldsymbol{\tau}\|_V}\\ &\le C_4\sup_{\boldsymbol{\tau}_h\in V_{h,\Gamma_S}}\frac{(v_h,\nabla\cdot\boldsymbol{\tau}_h)}{\|\boldsymbol{\tau}_h\|_V}\ (C_4:=C_1\tilde{C},\ \text{independently of } h).\end{aligned}$$

Proposition 3.2. *If* **(MVI)**$_h$ *admits a solution* $(\boldsymbol{\sigma}_h,u_h)\in\tilde{V}_h\times Q_h$, *then* $\mathring{\boldsymbol{\sigma}}_h:=\boldsymbol{\sigma}_h+\boldsymbol{\sigma}_{h,f}$ *satisfies* **(MVI)**$_h^\circ$. *Conversely, if* $\mathring{\boldsymbol{\sigma}}_h\in\mathring{\tilde{V}}_h$ *is a solution of* **(MVI)**$_h^\circ$, *then there exists a* $u_h\in Q_h$ *such that* $(\boldsymbol{\sigma}_h:=\mathring{\boldsymbol{\sigma}}_h-\boldsymbol{\sigma}_{h,f},u_h)$ *solves* **(MVI)**$_h$.

Proof. (1) Suppose $(\boldsymbol{\sigma}_h,u_h)\in\tilde{V}_h\times Q_h$ solves **(MVI)**$_h$. In view of $-\nabla\cdot\boldsymbol{\sigma}_h=\nabla\cdot\boldsymbol{\sigma}_{h,f}=f_h$ and $\boldsymbol{\sigma}_{h,f}=\mathbf{0}$ on Γ_S, we see that $\mathring{\boldsymbol{\sigma}}_h:=\boldsymbol{\sigma}_h+\boldsymbol{\sigma}_{h,f}\in\mathring{\tilde{V}}_h$. In addition, (43) is a direct result from (30a). Thus, $\mathring{\boldsymbol{\sigma}}_h$ solves **(MVI)**$_h^\circ$.

(2) Let $\mathring{\boldsymbol{\sigma}}_h\in\mathring{\tilde{V}}_h$ be a solution of **(MVI)**$_h^\circ$. Set $\boldsymbol{\sigma}_h:=\mathring{\boldsymbol{\sigma}}_h-\boldsymbol{\sigma}_{h,f}$. We find that $\boldsymbol{\sigma}_h\in\tilde{V}_h$ and $\nabla\cdot\boldsymbol{\sigma}_h=-\nabla\cdot\boldsymbol{\sigma}_{h,f}=-f_h$. By the inf-sup condition (44), we can find a unique $u_h\in Q_h$ such that

$$(u_h,\nabla\cdot\boldsymbol{\tau}_h)=-(\boldsymbol{\sigma}_h,\boldsymbol{\tau}_h)\quad(\forall\boldsymbol{\tau}_h\in V_{h,\Gamma_S}). \tag{46}$$

Moreover, for any $\boldsymbol{\tau}_h\in\tilde{V}_h$, there exists a $\boldsymbol{\xi}_h\in V_{h,\Gamma_S}$ such that

$$\nabla\cdot\boldsymbol{\xi}_h=\nabla\cdot\boldsymbol{\tau}_h+\nabla\cdot\boldsymbol{\sigma}_{h,f}. \tag{47}$$

Setting $\mathring{\boldsymbol{\tau}}_h:=\boldsymbol{\tau}_h+\boldsymbol{\sigma}_{h,f}-\boldsymbol{\xi}_h\in\mathring{\tilde{V}}_h$, we observe that: for any $\boldsymbol{\tau}_h\in\tilde{V}_h$,

$$\begin{aligned}(\boldsymbol{\sigma}_h,\boldsymbol{\tau}_h)&=(\boldsymbol{\sigma}_h,\mathring{\boldsymbol{\tau}}_h-\boldsymbol{\sigma}_{h,f}+\boldsymbol{\xi}_h)=(\boldsymbol{\sigma}_h,\mathring{\boldsymbol{\tau}}_h-\boldsymbol{\sigma}_{h,f})-(u_h,\nabla\cdot\boldsymbol{\xi}_h)\quad(\text{by (46)})\\ &=(\boldsymbol{\sigma}_h,\mathring{\boldsymbol{\tau}}_h-\boldsymbol{\sigma}_{h,f})-(u_h,\nabla\cdot(\boldsymbol{\tau}_h+\boldsymbol{\sigma}_{h,f}))\quad(\text{by (47)}),\end{aligned}$$

which yields

$$\begin{aligned}&(\boldsymbol{\sigma}_h,\boldsymbol{\tau}_h-\boldsymbol{\sigma}_h)=(\boldsymbol{\sigma}_h,\mathring{\boldsymbol{\tau}}_h-\boldsymbol{\sigma}_{h,f}-\boldsymbol{\sigma}_h)-(u_h,\nabla\cdot(\boldsymbol{\tau}_h+\boldsymbol{\sigma}_{h,f}))\\ =&(\mathring{\boldsymbol{\sigma}}_h-\boldsymbol{\sigma}_{h,f},\mathring{\boldsymbol{\tau}}_h-\mathring{\boldsymbol{\sigma}}_h)-(u_h,\nabla\cdot(\boldsymbol{\tau}_h-\boldsymbol{\sigma}_h))\quad(\text{by }\boldsymbol{\sigma}_h=\mathring{\boldsymbol{\sigma}}_h-\boldsymbol{\sigma}_{h,f}\text{ and }\boldsymbol{\sigma}_h=\mathring{\boldsymbol{\sigma}}_h-\boldsymbol{\sigma}_{h,f})\\ \ge&-(u_h,\nabla\cdot(\boldsymbol{\tau}_h-\boldsymbol{\sigma}_h))\quad(\text{by (43)}).\end{aligned}$$

Therefore, $(\boldsymbol{\sigma}_h,u_h)$ solves (30). □

Theorem 3.1 (*The Well-Posedness of* **(MVI)**$_h$). **(MVI)**$_h$ *admits a unique solution* $(\boldsymbol{\sigma}_h,u_h)\in\tilde{V}_h\times Q_h$ *with the estimate*

$$\|\boldsymbol{\sigma}_h\|_V+\|u_h\|_Q\le C\|f\|_Q. \tag{48}$$

Proof. According to Proposition 3.1, we only need to show the well-posedness of **(MVI)**$_h^\circ$. Notice that (43) is equivalent to

$$(\mathring{\boldsymbol{\sigma}}_h-\boldsymbol{\sigma}_{h,f},\mathring{\boldsymbol{\tau}}_h-\mathring{\boldsymbol{\sigma}}_h)_V\ge0\quad(\forall\mathring{\boldsymbol{\tau}}_h\in\mathring{\tilde{V}}_h). \tag{49}$$

Since $\mathring{\tilde{V}}_h$ is a closed convex subset of $\mathring{V}_h$, (49) admits a unique solution $\mathring{\boldsymbol{\sigma}}_h$, which is nothing but the projection of $\boldsymbol{\sigma}_{h,f}$ onto $\mathring{\tilde{V}}_h$. It remains to show the a-prior-estimate (48). Substituting $\boldsymbol{\tau}_h=\mathbf{0},2\boldsymbol{\sigma}_h$ into (30a) yields

$$\|\boldsymbol{\sigma}_h\|_Q^2=-(u_h,\nabla\cdot\boldsymbol{\sigma}_h)=(u_h,f_h)\le\|u_h\|_Q\|f_h\|_Q\quad(\text{by }\nabla\cdot\boldsymbol{\sigma}_h=-f_h). \tag{50}$$

Applying the inf-sup condition (44) to (46), we have $\|u_h\|_Q\le C_4\|\boldsymbol{\sigma}_h\|_Q$, which together with (50), results (48). □

For briefness, we set the notations:

$$\begin{aligned}&\boldsymbol{e}_{\boldsymbol{\sigma},h}:=\boldsymbol{\sigma}-\boldsymbol{\sigma}_h=(\boldsymbol{\sigma}-\Pi_h\boldsymbol{\sigma})+(\Pi_h\boldsymbol{\sigma}-\boldsymbol{\sigma}_h)=:\boldsymbol{\rho}_h+\boldsymbol{\theta}_h,\\ &e_{u,h}:=u-u_h,\quad e_{p,h}:=p-p_h.\end{aligned}$$

Theorem 3.2. *Let* $(\boldsymbol{\sigma},u)$ *and* $(\boldsymbol{\sigma}_h,u_h)$ *be the unique solutions of* **(MVI)** *and* **(MVI)**$_h$, *respectively. Let* p *and* p_h *satisfy* (5a) *and* (36a), *respectively. Assume that* $\boldsymbol{\sigma}\in H^1(\Omega)^d$, $\nabla\cdot\boldsymbol{\sigma}\in H^1(\Omega)$, $\boldsymbol{\sigma}\cdot\boldsymbol{n}|_{\Gamma_S}\in H^1(\Gamma_S)$, $u\in H^1(\Omega)$ *and* $p\in H^1(\Gamma_S)$. *Then, we have*

$$\|\boldsymbol{e}_{\boldsymbol{\sigma},h}\|_V+\|e_{u,h}\|_Q+\|e_{p,h}\|_\Lambda\le Ch. \tag{51}$$

Proof. Based on Propositions 2.2 and 3.1, we see **(MVI)** and **(MLM-I)** are equivalent, as well as **(MVI)**$_h$ and **(MLM-I)**$_h$. Therefore, by using (5a), (5b), (36a) and (36b), we derive

$$(\boldsymbol{e}_{\boldsymbol{\sigma},h},\boldsymbol{\tau}_h)+(e_{u,h},\nabla\cdot\boldsymbol{\tau}_h)-(e_{p,h},\boldsymbol{\tau}_h\cdot\boldsymbol{n})_{\Gamma_S}=0\qquad\forall\boldsymbol{\tau}_h\in V_h, \tag{52a}$$

$$(\nabla\cdot \boldsymbol{e}_{\boldsymbol{\sigma},h}, v_h) = 0 \qquad \forall v_h \in Q_h. \tag{52b}$$

Substituting $\boldsymbol{\tau}_h = \boldsymbol{\theta}_h$ into (52a) yields

$$\begin{aligned} 0 &= (\boldsymbol{e}_{\boldsymbol{\sigma},h}, \boldsymbol{\theta}_h) + (e_{u,h}, \nabla\cdot\boldsymbol{\theta}_h) - (e_{p,h}, \boldsymbol{\theta}_h\cdot\boldsymbol{n})_{\Gamma_S} \\ &= (\boldsymbol{\rho}_h + \boldsymbol{\theta}_h, \boldsymbol{\theta}_h) + ((u - P_h u) + (P_h u - u_h), \nabla\cdot\boldsymbol{\theta}_h) - (p - p_h, \boldsymbol{\theta}_h\cdot\boldsymbol{n})_{\Gamma_S} \\ &= (\boldsymbol{\rho}_h, \boldsymbol{\theta}_h) + \|\boldsymbol{\theta}_h\|_Q^2 + (P_h u - u_h, \nabla\cdot\boldsymbol{\theta}_h) - (p - p_h, \boldsymbol{\theta}_h\cdot\boldsymbol{n})_{\Gamma_S} \quad \text{(by (31))}. \end{aligned} \tag{53}$$

The last two terms of the right-hand side are calculated as follows:

$$\begin{aligned} (P_h u - u_h, \nabla\cdot\boldsymbol{\theta}_h) &= (P_h u - u_h, \nabla\cdot\boldsymbol{e}_{\boldsymbol{\sigma},h} - \nabla\cdot\boldsymbol{\rho}_h) = (P_h u - u_h, -\nabla\cdot\boldsymbol{\rho}_h) \quad \text{(by (52b))}, \\ -(p - p_h, \boldsymbol{\theta}_h\cdot\boldsymbol{n})_{\Gamma_S} &= -(p - p_h, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{\sigma}_h\cdot\boldsymbol{n})_{\Gamma_S} \\ &= -(p, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} + (p_h, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} + (p, \boldsymbol{\sigma}_h\cdot\boldsymbol{n})_{\Gamma_S} - (p_h, \boldsymbol{\sigma}_h\cdot\boldsymbol{n})_{\Gamma_S} \\ &\geq -(p, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} \quad \text{(by } p_h \geq 0, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n} \geq 0, p \geq 0, \boldsymbol{\sigma}_h\cdot\boldsymbol{n} \geq 0 \text{ and (41a))} \\ &= -(p, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} \quad \text{(by } (p, \boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} = 0) \\ &= -(p - P_{\Lambda_h} p, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} \quad \text{(by (32) and (33))}. \end{aligned}$$

Therefore, we obtain from (53) that

$$\begin{aligned} \|\boldsymbol{\theta}_h\|_Q^2 &\leq -(\boldsymbol{\rho}_h, \boldsymbol{\theta}_h) + (p - P_{\Lambda_h} p, \Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{\sigma}\cdot\boldsymbol{n})_{\Gamma_S} + (P_h u - u_h, \nabla\cdot\boldsymbol{\rho}_h) \\ &\leq \|\boldsymbol{\rho}_h\|_Q\|\boldsymbol{\theta}_h\|_Q + \|p - P_{\Lambda_h} p\|_\Lambda\|\Pi_h\boldsymbol{\sigma}\cdot\boldsymbol{n} - \boldsymbol{\sigma}\cdot\boldsymbol{n}\|_\Lambda + \|P_h u - u_h\|_Q\|\nabla\cdot\boldsymbol{\rho}_h\|_Q \\ &\leq Ch\|\boldsymbol{\sigma}\|_{H^1(\Omega)}\|\boldsymbol{\theta}_h\|_Q + Ch^2\|p\|_{H^1(\Gamma_S)}\|\boldsymbol{\sigma}\cdot\boldsymbol{n}\|_{H^1(\Gamma_S)} + \|P_h u - u_h\|_Q Ch\|\nabla\cdot\boldsymbol{\sigma}\|_{H^1(\Omega)}. \end{aligned} \tag{54}$$

Meanwhile, it follows form (52a) that: for all $\boldsymbol{\tau}_h \in V_{h,\Gamma_S}$, (by (31) and $\nabla\cdot\boldsymbol{\tau}_h \in Q_h$)

$$(P_h u - u_h, \nabla\cdot\boldsymbol{\tau}_h) = -(\boldsymbol{e}_{\boldsymbol{\sigma},h}, \boldsymbol{\tau}_h) - (u - P_h u, \nabla\cdot\boldsymbol{\tau}_h) = -(\boldsymbol{\rho}_h + \boldsymbol{\theta}_h, \boldsymbol{\tau}_h). \tag{55}$$

Applying the inf-sup condition (44) to (55), we get

$$\|P_h u - u_h\|_Q \leq C(\|\boldsymbol{\rho}_h\|_Q + \|\boldsymbol{\theta}_h\|_Q) \leq Ch\|\boldsymbol{\sigma}\|_{H^1(\Omega)} + C\|\boldsymbol{\theta}_h\|_Q. \tag{56}$$

Inserting (56) into the right-hand side of (54) results

$$\|\boldsymbol{\theta}_h\|_Q \leq Ch(\|\boldsymbol{\sigma}\|_{H^1(\Omega)} + \|\nabla\cdot\boldsymbol{\sigma}\|_{H^1(\Omega)} + \|p\|_{H^1(\Gamma_S)} + \|\boldsymbol{\sigma}\cdot\boldsymbol{n}\|_{H^1(\Gamma_S)}).$$

On the other hand, it follows from (52b) that

$$(\nabla\cdot\boldsymbol{\theta}_h, v_h) = -(\nabla\cdot\boldsymbol{\rho}_h, v_h) \quad \forall v_h \in Q_h,$$

which implies

$$\|\nabla\cdot\boldsymbol{\theta}_h\|_Q \leq \|\nabla\cdot\boldsymbol{\rho}_h\|_Q \leq Ch\|\nabla\cdot\boldsymbol{\sigma}\|_{H^1(\Omega)}.$$

By the inf-sup condition (40) and (52a),

$$\begin{aligned} \|P_{\Lambda_h} p - p_h\|_\Lambda &\leq C_3 \sup_{\boldsymbol{\tau}_h\in V_h} \frac{(P_{\Lambda_h} p - p_h, \boldsymbol{\tau}_h\cdot\boldsymbol{n})_{\Gamma_S}}{\|\boldsymbol{\tau}_h\|_V} \\ &= C_3 \sup_{\boldsymbol{\tau}_h\in V_h} \frac{-(p - P_{\Lambda_h} p, \boldsymbol{\tau}_h\cdot\boldsymbol{n}) + (\boldsymbol{e}_{\boldsymbol{\sigma},h}, \boldsymbol{\tau}_h) + (e_{u,h}, \nabla\cdot\boldsymbol{\tau}_h)}{\|\boldsymbol{\tau}_h\|_V} \\ &= C_3 \sup_{\boldsymbol{\tau}_h\in V_h} \frac{0 + (\boldsymbol{\rho}_h + \boldsymbol{\theta}_h, \boldsymbol{\tau}_h) + (P_h u - u_h, \nabla\cdot\boldsymbol{\tau}_h)}{\|\boldsymbol{\tau}_h\|_V} \quad \text{(by (35) and (31))} \\ &\leq C_3(\|\boldsymbol{\rho}_h\|_Q + \|\boldsymbol{\theta}_h\|_Q + \|P_h u - u_h\|_Q). \end{aligned} \tag{57}$$

A combination of (56)–(57) results (51). □

3.2. The active/inactive set method

To solve the mixed variational inequality, we propose two Active/Inactive set algorithms based on the discrete mixed Lagrange multiplier formulations **(MLM-I)**$_h$ and **(MLM-II)**$_h$, respectively.

Note that (36c) of **(MLM-I)**$_h$ can be written into

$$(\boldsymbol{\tau}_h\cdot\boldsymbol{n} - \boldsymbol{\sigma}_h\cdot\boldsymbol{n}, -\rho p_h + \boldsymbol{\sigma}_h\cdot\boldsymbol{n} - \boldsymbol{\sigma}_h\cdot\boldsymbol{n})_{\Gamma_S} \leq 0 \quad (\forall\rho > 0),$$

which is equivalent to

$$\boldsymbol{\sigma}_h\cdot\boldsymbol{n} = P_{\tilde{\Lambda}_h}(\boldsymbol{\sigma}_h\cdot\boldsymbol{n} - \rho p_h) = \max(0, \boldsymbol{\sigma}_h\cdot\boldsymbol{n} - \rho p_h). \tag{58}$$

The Active/Inactive set method (Algorithm 1) presented below is based on the iteration

$$\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = P_{\check{\Lambda}_h}(\boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n} - \rho p_h^{(n-1)}). \tag{59}$$

Algorithm 1 The Active/Inactive set method based on **(MLM-I)**$_h$

(Step 1) Choose a $\rho > 0$. Find $(\boldsymbol{\sigma}_h^{(0)}, u_h^{(0)}) \in V_h \times Q_h$ such that

$$\begin{aligned} &\boldsymbol{\sigma}_h^{(0)} \cdot \boldsymbol{n} = 0 && \text{on } \Gamma_S, \\ &(\boldsymbol{\sigma}_h^{(0)}, \boldsymbol{\tau}_h) + (u_h^{(0)}, \nabla \cdot \boldsymbol{\tau}_h) = 0 && \forall \boldsymbol{\tau}_h \in V_h, \\ &(\nabla \cdot \boldsymbol{\sigma}_h^{(0)}, v_h) + (f_h, v_h) = 0 && \forall v_h \in Q_h. \end{aligned}$$

Set $p_h^{(0)} = 0$ and $n = 1$.

(Step 2) Decompose Γ_S into two subsets:

$$\begin{aligned} &\Gamma_S^{1,(n)} := \{e \in \Gamma_S : \boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n} - \rho p_h^{(n-1)} < 0 \text{ on } e\} && \text{(The active set)}, \\ &\Gamma_S^{2,(n)} := \Gamma_S \backslash \Gamma_S^{1,(n)} && \text{(The inactive set)}. \end{aligned}$$

(Step 3) Find $(\boldsymbol{\sigma}_h^{(n)}, u_h^{(n)}) \in V_h \times Q_h$ such that

$$\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = 0 \qquad \text{on } \Gamma_S^{1,(n)}, \tag{60a}$$

$$(\boldsymbol{\sigma}_h^{(n)}, \boldsymbol{\tau}_h) + (u_h^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) = 0 \qquad \forall \boldsymbol{\tau}_h \in V_{h,\Gamma_S^{1,(n)}}, \tag{60b}$$

$$(\nabla \cdot \boldsymbol{\sigma}_h^{(n)}, v_h) + (f_h, v_h) = 0 \qquad \forall v_h \in Q_h. \tag{60c}$$

(Step 4) Find $p_h^{(n)} \in \Lambda_h$ such that

$$(p_h^{(n)}, \boldsymbol{\tau}_h \cdot \boldsymbol{n})_{\Gamma_S} = (\boldsymbol{\sigma}_h^{(n)}, \boldsymbol{\tau}_h) + (u_h^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) \quad \forall \boldsymbol{\tau}_h \in V_h. \tag{61}$$

(Step 5) $n \to n+1$. Iterate (Step2)-(Step4) until convergence.

Remark 3.6. In (Step 3) of Algorithm 1, the projection $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = \max(0, \boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n} - \rho p_h^{(n-1)})$ is carried out on the active set $\Gamma_S^{1,(n)}$ only (see Eq. (60a)), while on the inactive set $\Gamma_S^{2,(n)}$ we do nothing. Actually, it follows from (60b) and (61) that

$$p_h^{(n)} = 0 \text{ on } \Gamma_S^{2,(n)}. \tag{62}$$

Proposition 3.3. *Assume that after n_0 iteration steps, the active(also inactive) sets becomes invariant, i.e., $\Gamma_S^{1,(n_0+1)} = \Gamma_S^{1,(n_0)}$ (also $\Gamma_S^{2,(n_0+1)} = \Gamma_S^{2,(n_0)}$). Then Algorithm 1 converges, that is,*

$$(\boldsymbol{\sigma}_h^{(n+1)}, u_h^{(n+1)}, p_h^{(n+1)}) = (\boldsymbol{\sigma}_h^{(n)}, u_h^{(n)}, p_h^{(n)}) \quad (\forall n \geq n_0).$$

Moreover, we have $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} \geq 0$, $p_h^{(n)} \geq 0$, $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} p_h^{(n)} = 0$ on Γ_S $(\forall n \geq n_0)$.

Remark 3.7. The convergence analysis of the Active/Inactive set method usually relies on the assumption of the invariant active (and inactive) set after a couple of iterations. We refer to [37, Theorem 4.1] and [40, Theorem 3.4] for details. In fact, this assumption cannot be prescribed. Nevertheless, in our numerical experiments, the algorithm performed well, and the iteration stopped after only a few steps (refer to Figs. 4, 7, and 10 for Algorithms 1 and 2).

Proof of Proposition 3.3. Under the assumption $\Gamma_S^{1,(n+1)} = \Gamma_S^{1,(n)}$, it follows from (60) that $(\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}, u_h^{(n+1)} - u_h^{(n)})$ satisfies:

$$(\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}) \cdot \boldsymbol{n} = 0 \qquad \text{on } \Gamma_S^{1,(n)}, \tag{63a}$$

$$(\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}, \boldsymbol{\tau}_h) + (u_h^{(n+1)} - u_h^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) = 0 \qquad \forall \boldsymbol{\tau}_h \in V_{h,\Gamma_S^{1,(n)}}, \tag{63b}$$

$$\nabla \cdot (\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}) = 0 \qquad \text{in } \Omega. \tag{63c}$$

Substituting $\boldsymbol{\tau}_h = \boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)} \in V_{h,\Gamma_S^{1,(n)}}$ into (63b), and using (63c), we obtain

$$\|\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}\|_Q^2 + 0 = 0,$$

i.e., $\boldsymbol{\sigma}_h^{(n+1)} = \boldsymbol{\sigma}_h^{(n)}$. Moreover, $u_h^{(n+1)} = u_h^{(n)}$ follows from (63b) and the inf-sup condition (44). From (61) and the inf-sup condition (40), we get $p_h^{(n+1)} = p_h^{(n)}$.

In view of $\Gamma_S^{1,(n+1)} = \Gamma_S^{1,(n)}, \Gamma_S^{2,(n+1)} = \Gamma_S^{2,(n)}$, (60a) and (62), we have

$$\begin{aligned} &\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = 0, \qquad p_h^{(n)} > \rho^{-1}\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = 0 \quad \text{on } \Gamma_S^{1,(n)}, \\ &p_h^{(n)} = 0, \qquad \boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} \geq \rho p_h^{(n)} = 0 \quad \text{on } \Gamma_S^{2,(n)}, \end{aligned}$$

which implies $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} \geq 0, p_h^{(n)} \geq 0, \boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} p_h^{(n)} = 0$ on Γ_S. □

From (37c), we see that, for arbitrary $\rho > 0$,

$$(q_h - p_h, -\rho\boldsymbol{\sigma}_h \cdot \boldsymbol{n} + p_h - p_h) \leq 0 \quad \forall q_h \in \tilde{\Lambda}_h,$$

which is equivalent to

$$p_h = P_{\tilde{\Lambda}_h}(p_h - \rho\boldsymbol{\sigma}_h \cdot \boldsymbol{n}) = \max(0, p_h - \rho\boldsymbol{\sigma}_h \cdot \boldsymbol{n}). \tag{64}$$

Algorithm 2 The Active/Inactive set method based on **(MLM-II)**$_h$

(Step 1) Same to the Algorithm 1.

(Step 2) Decompose Γ_S into two subsets:

$$\begin{aligned} &\Gamma_S^{1,(n)} := \{e \in \Gamma_S : p_h^{(n-1)} - \rho\sigma_h^{(n-1)} \cdot \boldsymbol{n} < 0 \text{ on } e\} \qquad \text{(The active set)}, \\ &\Gamma_S^{2,(n)} := \Gamma_S \backslash \Gamma_S^{1,(n)} \qquad \text{(The inactive set)}. \end{aligned}$$

(Step 3) Find $(\boldsymbol{\sigma}_h^{(n)}, u_h^{(n)}) \in V_h \times Q_h$ such that

$$\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = 0 \qquad \text{on } \Gamma_S^{2,(n)}, \tag{65a}$$

$$(\boldsymbol{\sigma}_h^{(n)}, \boldsymbol{\tau}_h) + (u_h^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) = 0 \qquad \forall \boldsymbol{\tau}_h \in V_{h,\Gamma_S^{2,(n)}}, \tag{65b}$$

$$(\nabla \cdot \boldsymbol{\sigma}_h^{(n)}, v_h) + (f_h, v_h) = 0 \qquad \forall v_h \in Q_h. \tag{65c}$$

(Step 4) Find $p_h^{(n)} \in \Lambda_h$ such that

$$(p_h^{(n)}, \boldsymbol{\tau}_h \cdot \boldsymbol{n})_{\Gamma_S} = (\boldsymbol{\sigma}_h^{(n)}, \boldsymbol{\tau}_h) + (u_h^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) \quad \forall \boldsymbol{\tau}_h \in V_h. \tag{66}$$

(Step 5) $n \to n+1$. Iterate (Step2)-(Step4) until convergence.

Remark 3.8. According to (65b) and (66), it can be observed that $p_h^{(n)} = 0$ on $\Gamma_S^{1,(n)}$, which is the same as performing the projection $p_h^{(n)} = P_{\tilde{\Lambda}_h}(p_h^{(n-1)} - \rho\boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n})$ on $\Gamma_S^{1,(n)}$.

Proposition 3.4. *Assume that after n_0 iterations, the active (and inactive) sets are not changed, i.e., $\Gamma_S^{1,(n_0+1)} = \Gamma_S^{1,(n_0)}$ (and $\Gamma_S^{2,(n_0+1)} = \Gamma_S^{2,(n_0)}$). Then, we have: for all $n \geq n_0$*

$$(\boldsymbol{\sigma}_h^{(n+1)}, u_h^{(n+1)}, p_h^{(n+1)}) = (\boldsymbol{\sigma}_h^{(n)}, u_h^{(n)}, p_h^{(n)}). \tag{67}$$

Moreover, we have $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} \geq 0$, $p_h^{(n)} \geq 0$ and $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} p_h^{(n)} = 0$ on Γ_S.

Proof. Suppose $\Gamma_S^{1,(n+1)} = \Gamma_S^{1,(n)}$. Then, it follows from (65) that

$$\begin{aligned} &(\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}) \cdot \boldsymbol{n} = 0 \qquad \text{on } \Gamma_S^{1,(n)}, \\ &(\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}, \boldsymbol{\tau}_h) + (u_h^{(n+1)} - u_h^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) = 0 \qquad \forall \boldsymbol{\tau}_h \in V_{h,\Gamma_S^{1,(n)}}, \\ &\nabla \cdot (\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}) = 0 \qquad \text{in } \Omega, \end{aligned}$$

which implies $\boldsymbol{\sigma}_h^{(n+1)} = \boldsymbol{\sigma}_h^{(n)}$. By using inf-sup condition (44), we observe $u_h^{(n+1)} = u_h^{(n)}$. Together with (66) and inf-sup condition (40), we assert $p_h^{(n+1)} = p_h^{(n)}$.

According to Remark 3.8 and $\Gamma_S^{1,(n+1)} = \Gamma_S^{1,(n)}$, we see that

$$p^{(n)} = 0, \quad \boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} > \rho^{-1} p^{(n)} = 0 \text{ on } \Gamma_S^{1,(n)}.$$

On the other hand, because of $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = 0$ on $\Gamma_S^{2,(n)}$ and $\Gamma_S^{2,(n+1)} = \Gamma_S^{2,(n)}$, we have

$$p_h^{(n)} \geq \rho\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} = 0 \text{ on } \Gamma_S^{2,(n)}.$$

Hence, we conclude (67) and $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} \geq 0$, $p_h^{(n)} \geq 0$ and $\boldsymbol{\sigma}_h^{(n)} \cdot \boldsymbol{n} p_h^{(n)} = 0$ on Γ_S $(n \geq n_0)$. □

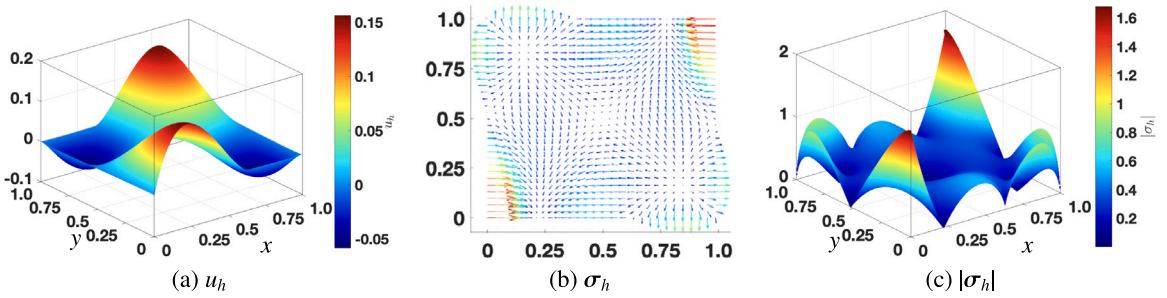


Fig. 1. Example 1: The numerical solution (σ_h, u_h).

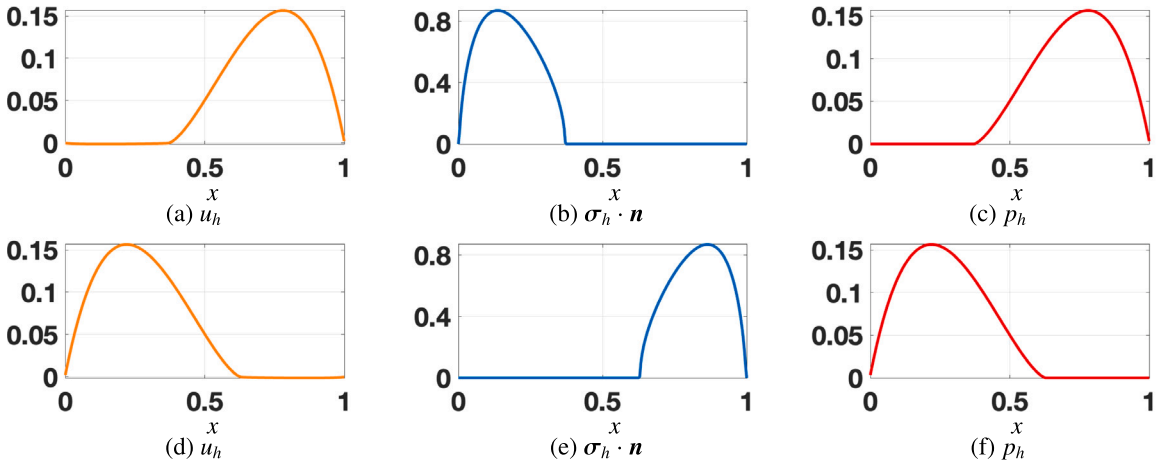


Fig. 2. Example 1: $(u_h, \sigma_h \cdot \boldsymbol{n}, p_h)$ on Γ_{S1} (top) and Γ_{S2} (bottom) of $\textbf{(MLM-I)}_h$ by using Algorithm 1.

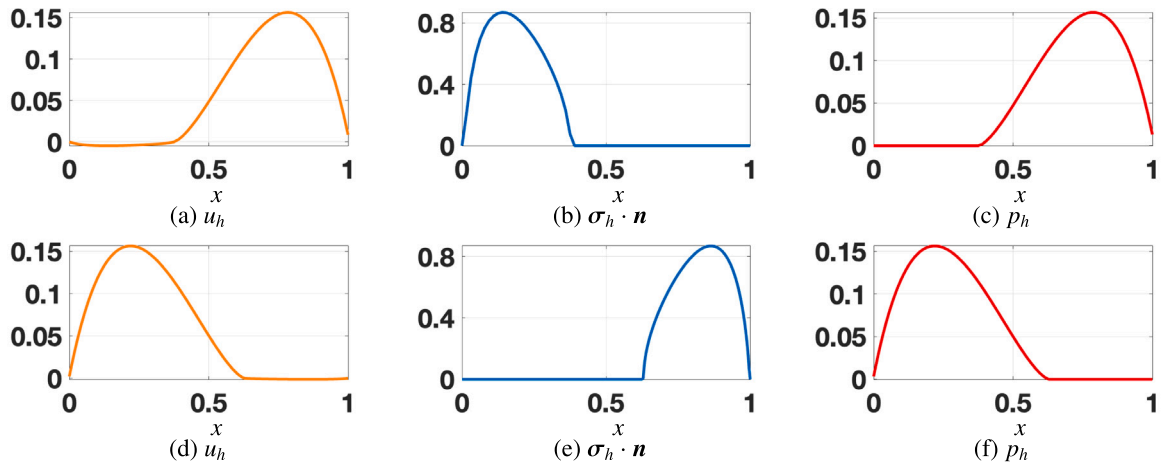


Fig. 3. Example 1: $(u_h, \sigma_h \cdot \boldsymbol{n}, p_h)$ on Γ_{S1} (top) and Γ_{S2} (bottom) of $\textbf{(MLM-II)}_h$ by using Algorithm 2.

Remark 3.9. The assumption that the active sets are invariant after a couple of iterations cannot be prescribed. In Appendix, we provide two Uzawa algorithms based on the projection formulations (59) and (64), respectively. The convergence of the Uzawa methods can be guaranteed by choosing a sufficiently small parameter ρ. However, in practice, the Uzawa methods usually take much more iterations than the Active/Inactive set methods.

Table 1
Example 1: The errors $\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$ and $\|u_{ref}-u_h\|_Q$ for Algorithm 1.

h	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
$\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$	1.65×10^{0}	7.95×10^{-1}	3.83×10^{-1}	1.78×10^{-1}	7.64×10^{-2}
Order	–	1.05	1.06	1.10	1.22
$\|u_{ref}-u_h\|_Q$	4.65×10^{-2}	2.47×10^{-2}	1.21×10^{-2}	5.68×10^{-3}	2.44×10^{-3}
Order	–	0.91	1.03	1.10	1.22

Table 2
Example 1: The errors $\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$ and $\|u_{ref}-u_h\|_Q$ for Algorithm 2.

h	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
$\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$	1.65×10^{0}	7.95×10^{-1}	3.83×10^{-1}	1.78×10^{-1}	7.64×10^{-2}
Order	–	1.05	1.06	1.10	1.22
$\|u_{ref}-u_h\|_Q$	4.65×10^{-2}	2.47×10^{-2}	1.21×10^{-2}	5.68×10^{-3}	2.44×10^{-3}
Order	–	0.91	1.03	1.10	1.22

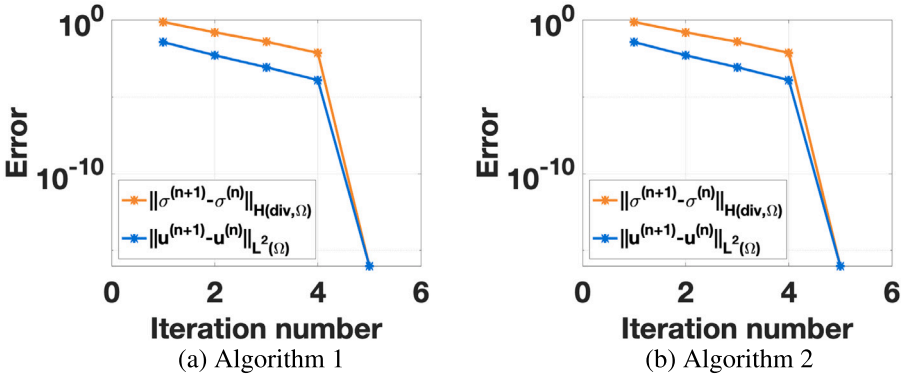


Fig. 4. Example 1: The iteration errors $\|\boldsymbol{\sigma}_h^{(n+1)}-\boldsymbol{\sigma}_h^{(n)}\|_V$ and $\|u_h^{(n+1)}-u_h^{(n)}\|_Q$ by Algorithms 1 and 2 with $h=2^{-6}$.

4. Numerical experiments

To investigate the rates of convergence for discretization and iteration algorithms, we conducted three numerical examples using various geometric domains.

4.1. Example 1

We consider a square domain $\Omega=(0,1)^2$, with $\Gamma_S=\Gamma_{S_1}\cup\Gamma_{S_2}=(0,1)\times\{1\}\cup(0,1)\times\{0\}$, and $\Gamma_D=\partial\Omega\setminus\Gamma_S$. Set $f=50(\frac{1}{2}-x)(\frac{1}{2}-y)$.

The simulation results for Algorithms 1 and 2 are almost identical, so we only display the numerical solution $(\boldsymbol{\sigma}_h,u_h)$ of Algorithm 1 in Fig. 1. Furthermore, we plot u_h, $\boldsymbol{\sigma}_h\cdot\boldsymbol{n}$ and p_h on Γ_S (see Figs. 2 and 3 for Algorithms 1 and 2). We observe that $u_h|_{\Gamma_S}\approx p_h$, and the numerical results accurately approximate the Signorini conditions $\boldsymbol{\sigma}_h\cdot\boldsymbol{n}\geq0$, $p_h\geq0$ and $(\boldsymbol{\sigma}_h\cdot\boldsymbol{n},p_h)=0$.

To verify the theoretical convergence rates $O(h)$ Theorem 3.2, we compute the errors $\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$ and $\|u_{ref}-u_h\|_Q$ for $h=2^{-2}$, 2^{-3}, 2^{-4}, 2^{-5}, 2^{-6} where $(\boldsymbol{\sigma}_{ref},u_{ref})$ is the reference solution with the mesh size $h=2^{-8}$ (see Tables 1 and 2). The experiment shows first-order convergence. Furthermore, we explore the efficiency of Algorithms 1 and 2. We compute the errors $\|\boldsymbol{\sigma}_h^{(n+1)}-\boldsymbol{\sigma}_h^{(n)}\|_V$ and $\|u_h^{(n+1)}-u_h^{(n)}\|_Q$ $(n=1,2,\ldots)$ at each step and we observe a rapid decrease in them. This is shown in Fig. 4(a) and (b) for $h=2^{-6}$. The outcomes are relatively consistent for other choices of h.

4.2. Example 2

Set an L-shaped region $\Omega=(0,2)^2\setminus[1,2]^2$ with $\Gamma_S=\Gamma_{S_1}\cup\Gamma_{S_2}=(0,2)\times\{0\}\cup\{0\}\times(0,2)$ and $\Gamma_D=\partial\Omega\setminus\Gamma_S$, and $f=25(\frac{1}{2}-x)(\frac{1}{2}-y)$.

The numerical solution is plotted in Fig. 5 which is computed by Algorithm 1, as well as the figures of u_h, $\boldsymbol{\sigma}\cdot_h\boldsymbol{n}$ and p_h on Γ_{S1} and Γ_{S2} are displayed in Fig. 6.

As in Example 1, we calculate the errors $\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$ and $\|u_{ref}-u_h\|_Q$ for various mesh sizes h in Tables 3 and 4, which indicate $O(h)$-convergence for the error $\|u_{ref}-u_h\|_Q$. Since a singularity occurs at the vertex $(1,1)$ (see Fig. 5(c)), we observe less

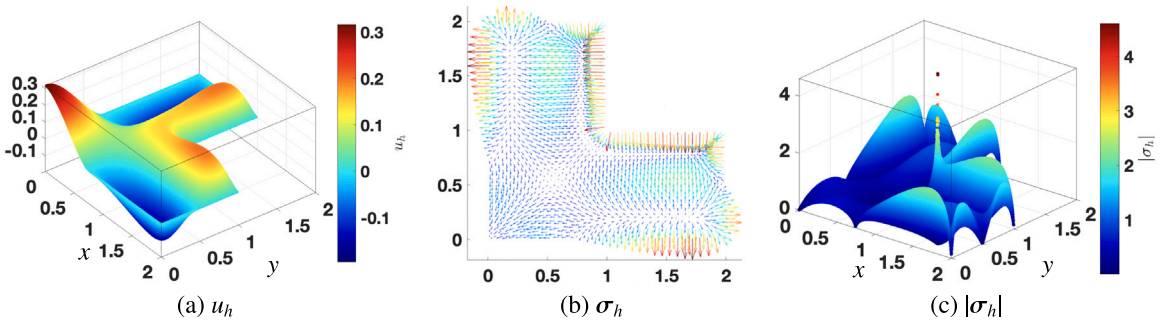


Fig. 5. Example 2: The numerical solution $(\boldsymbol{\sigma}_h, u_h)$.

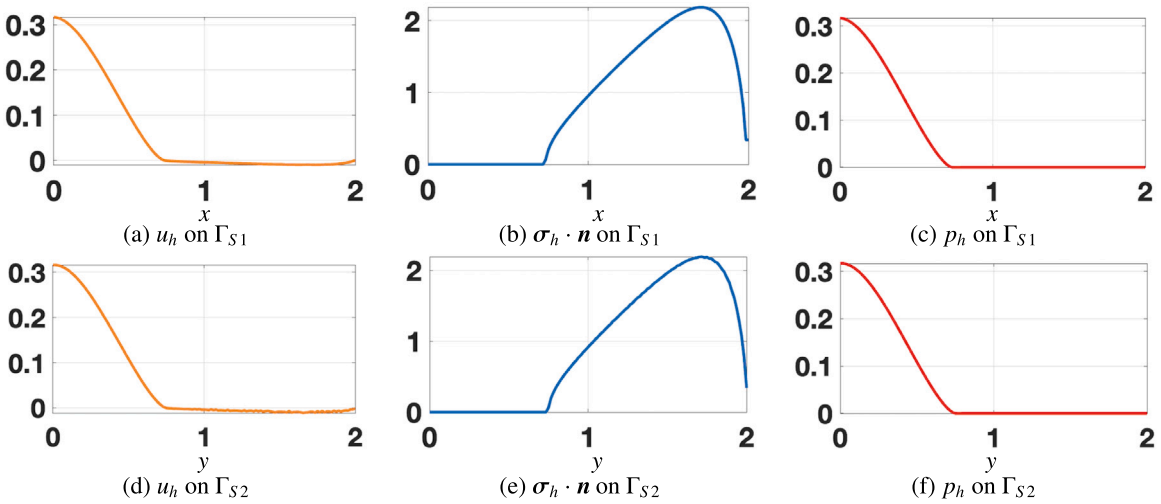


Fig. 6. Example 2: $(u_h, \boldsymbol{\sigma}_h \cdot \boldsymbol{n}, p_h)$ on Γ_{S1} (top) and Γ_{S2} (bottom) by using Algorithm 1.

Table 3
Example 2: The errors $\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$ and $\|u_{ref} - u_h\|_Q$ for Algorithm 1.

h	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
$\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$	2.17×10^{0}	1.02×10^{0}	$4.79s \times 10^{-1}$	2.52×10^{-1}	1.39×10^{-1}
Order	–	1.14	1.30	0.86	0.86
$\|u_{ref} - u_h\|_Q$	1.14×10^{-1}	5.39×10^{-2}	2.29×10^{-2}	1.03×10^{-2}	5.11×10^{-3}
Order	–	1.13	1.48	1.06	1.02

Table 4
Example 2: The errors $\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$ and $\|u_{ref} - u_h\|_Q$ for Algorithm 2.

h	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
$\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$	2.17×10^{0}	1.02×10^{0}	$4.79s \times 10^{-1}$	2.52×10^{-1}	1.39×10^{-1}
Order	–	1.14	1.30	0.86	0.86
$\|u_{ref} - u_h\|_Q$	1.14×10^{-1}	5.39×10^{-2}	2.29×10^{-2}	1.03×10^{-2}	5.11×10^{-3}
Order	–	1.13	1.48	1.06	1.02

than first-order convergence for the error $\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$. Moreover, the iteration errors $\|\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}\|_V$ and $\|u_h^{(n+1)} - u_h^{(n)}\|_Q$ with $h = 2^{-3}$ by Algorithms 1 and 2 are shown in Fig. 7(a) and (b). It seems that the Active/Inactive set method performs well.

4.3. Example 3

We conduct a test on the numerical schemes by solving a problem involving the Signorini condition on a curved boundary. We consider a half-disk domain $\Omega = \{(x, y) : x^2 + y^2 < 1, y < 0\}$ with $\Gamma_S = \{(x, -\sqrt{1 - x^2}) : x \in (-1, 1)\}$ and $\Gamma_D = \partial\Omega \setminus \Gamma_S$. Set $f = 5x$.

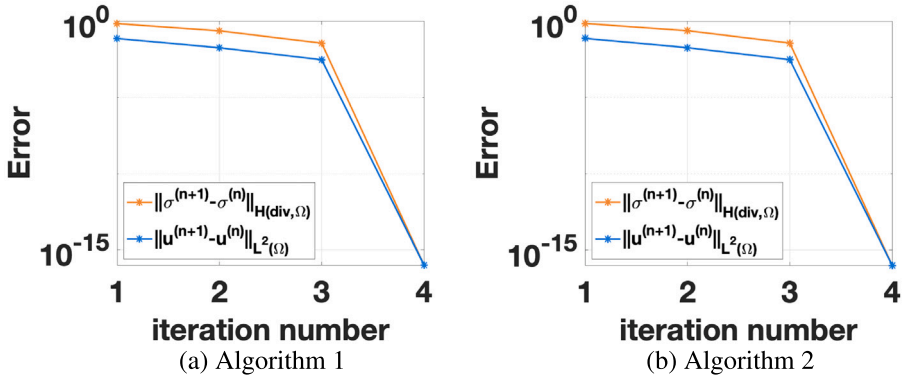


Fig. 7. Example 2: The iteration errors $\|\boldsymbol{\sigma}_h^{(n+1)} - \boldsymbol{\sigma}_h^{(n)}\|_V$ and $\|u_h^{(n+1)} - u_h^{(n)}\|_Q$ by Algorithms 1 and 2 with $h = 2^{-3}$.

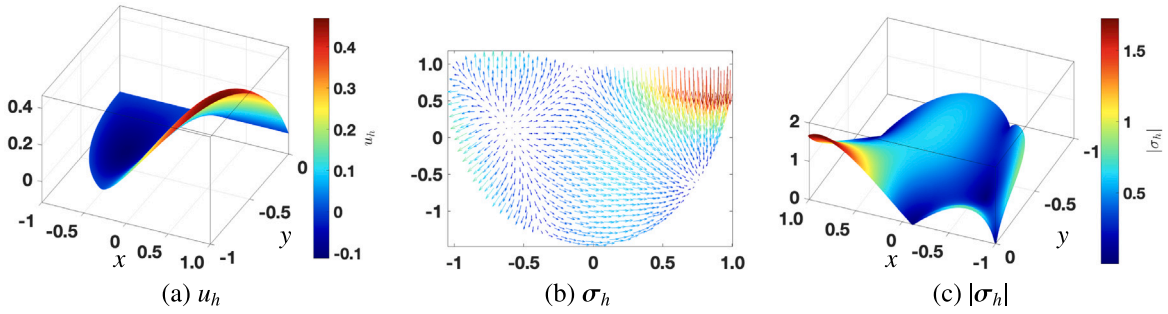


Fig. 8. Example 3: The numerical solution $(\boldsymbol{\sigma}_h, u_h)$.

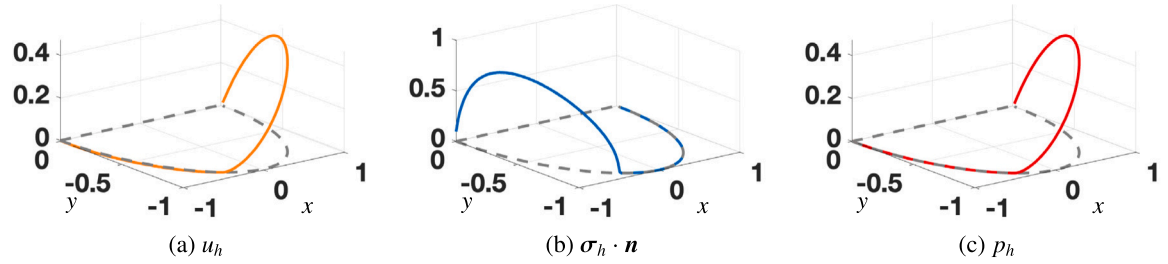


Fig. 9. Example 3: $(u_h, \boldsymbol{\sigma}_h \cdot \boldsymbol{n}, p_h)$ on Γ_S by using Algorithm 1.

Table 5
Example 3: The errors $\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$ and $\|u_{ref} - u_h\|_Q$ for Algorithm 1.

h	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
$\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$	3.11×10^{-1}	1.61×10^{-1}	7.20×10^{-2}	3.31×10^{-2}	1.60×10^{-2}
Order	–	0.97	1.27	1.03	1.17
$\|u_{ref} - u_h\|_Q$	5.28×10^{-2}	2.54×10^{-2}	1.11×10^{-2}	5.09×10^{-3}	2.52×10^{-3}
Order	–	1.07	1.32	1.03	1.12

The numerical solution of Algorithm 1 is depicted in Fig. 8, while Fig. 9 shows the plots of u_h, $\boldsymbol{\sigma}_h \cdot \boldsymbol{n}$ and p_h on Γ_S which satisfies the Signorini conditions.

Similar to Example 1, Tables 5 and 6 displays the $O(h)$-convergence of $\|\boldsymbol{\sigma}_{ref} - \boldsymbol{\sigma}_h\|_V$ and $\|u_{ref} - u_h\|_Q$. Under $h = 2^{-3}$, the iteration errors are shown in Fig. 10(a) and (b). This example shows that the mixed formulation is effective even with curved contact boundaries.

Table 6
Example 3: The errors $\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$ and $\|u_{ref}-u_h\|_Q$ for Algorithm 2.

h	2^{-3}	2^{-4}	2^{-5}	2^{-6}	2^{-7}
$\|\boldsymbol{\sigma}_{ref}-\boldsymbol{\sigma}_h\|_V$	3.11×10^{-1}	1.61×10^{-1}	7.20×10^{-2}	3.31×10^{-2}	1.60×10^{-2}
Order	–	0.97	1.27	1.03	1.17
$\|u_{ref}-u_h\|_Q$	5.28×10^{-2}	2.54×10^{-2}	1.11×10^{-2}	5.09×10^{-3}	2.52×10^{-3}
Order	–	1.07	1.32	1.03	1.12

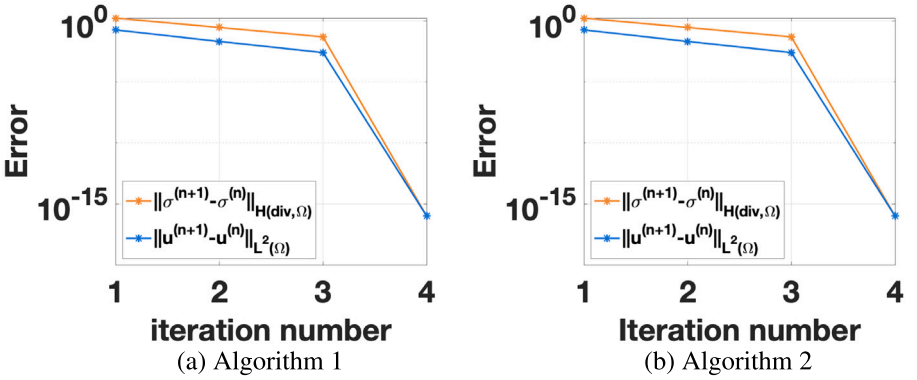


Fig. 10. Example 3: The iteration errors $\|\boldsymbol{\sigma}_h^{(n+1)}-\boldsymbol{\sigma}_h^{(n)}\|_V$ and $\|u_h^{(n+1)}-u_h^{(n)}\|_Q$ by Algorithms 1 and 2 with $h=2^{-3}$.

Data availability

Data will be made available on request.

Appendix. The Uzawa algorithm based on (MLM-I)$_h$ and (MLM-II)$_h$

We utilize the projection formulas (58) and (64) for the design of the Uzawa algorithms, and we have proved the convergence.

Algorithm 3 The Uzawa method based on **(MLM-I)**$_h$

(Step 1) Choose the parameter $\rho>0$. Set $\boldsymbol{\sigma}_h^{(0)}\cdot\boldsymbol{n}=0$, $p_h^{(0)}=0$ and $n=1$.

(Step 2) Find $(\boldsymbol{\sigma}_h^{(n)},u_h^{(n)})\in V_h\times Q_h$ such that

$$\boldsymbol{\sigma}_h^{(n)}\cdot\boldsymbol{n}=P_{\tilde{\Lambda}_h}(\boldsymbol{\sigma}_h^{(n-1)}\cdot\boldsymbol{n}-\rho p_h^{(n-1)}) \quad \text{on } \Gamma_S, \tag{A.1a}$$

$$(\boldsymbol{\sigma}_h^{(n)},\boldsymbol{\tau}_h)+(u_h^{(n)},\nabla\cdot\boldsymbol{\tau}_h)=0 \quad \forall\boldsymbol{\tau}_h\in V_{h,\Gamma_S}, \tag{A.1b}$$

$$(\nabla\cdot\boldsymbol{\sigma}_h^{(n)},v_h)+(f_h,v_h)=0 \quad \forall v_h\in Q_h. \tag{A.1c}$$

(Step 3) Find $p_h^{(n)}\in\Lambda_h$ such that

$$(p_h^{(n)},\boldsymbol{\tau}_h\cdot\boldsymbol{n})_{\Gamma_S}=(\boldsymbol{\sigma}_h^{(n)},\boldsymbol{\tau}_h)+(u_h^{(n)},\nabla\cdot\boldsymbol{\tau}_h)\quad\forall\boldsymbol{\tau}_h\in V_h. \tag{A.2}$$

(Step 4) $n\to n+1$. Iterate (Step2) (Step3) until convergence.

Remark A.1. Let C_{T1} be the optimal constant of the following discrete trace inequality

$$\|\boldsymbol{\tau}_h\cdot\boldsymbol{n}\|_\Lambda\le C_{T1}\|\boldsymbol{\tau}_h\|_V\quad(\forall\boldsymbol{\tau}_h\in V_h). \tag{A.3}$$

By using the discrete inverse trace inequality, there is a constant C independent of h such that,

$$\|\boldsymbol{\tau}_h\cdot\boldsymbol{n}\|_\Lambda\le Ch^{-\frac{1}{2}}\|\boldsymbol{\tau}_h\|_Q\le Ch^{-\frac{1}{2}}\|\boldsymbol{\tau}_h\|_V\quad(\forall\boldsymbol{\tau}_h\in V_h),$$

where implies that C_{T1} has an upper bound, i.e., $C_{T1}\le Ch^{-\frac{1}{2}}$.

A combination of (39c) and (A.3) shows that

$$C_{T1}^{-1}\|\boldsymbol{\tau}_h\cdot\boldsymbol{n}\|_\Lambda\le\|\boldsymbol{\tau}_h\|_V\le C_3\|\boldsymbol{\tau}_h\cdot\boldsymbol{n}\|_\Lambda\quad(\forall\boldsymbol{\tau}_h\in V_h). \tag{A.4}$$

To be brief, we set $\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} := \boldsymbol{\sigma}_h - \boldsymbol{\sigma}_h^{(n)}$, $e_p := p_h - p_h^{(n)}$, $e_u^{(n)} := u_h - u_h^{(n)}$.

Proposition A.1. *Assume that ρ is sufficiently small such that*

$$0 < \rho < 2/(C_3^2(1+C_4)^2). \tag{A.5}$$

Then $(\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}, e_u^{(n)}, e_p^{(n)}) \to (\mathbf{0}, 0, 0)$ as $n \to \infty$. In addition, if $1 - (2\rho - \rho^2 C_3^2(1+C_4)^2)C_{T1}^{-2} < 1$, then we have

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V^2 \le C_3^2(1 - (2\rho - \rho^2 C_3^2(1+C_4)^2)C_{T1}^{-2})^n \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(0)} \cdot \boldsymbol{n}\|_\Lambda^2, \tag{A.6a}$$

$$\|e_u^{(n)}\|_Q \le C_4 \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V, \tag{A.6b}$$

$$\|e_p^{(n)}\|_\Lambda \le C_3(1+C_4)\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V. \tag{A.6c}$$

Proof. Subtracting (A.2) from (36a) results

$$(\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}, \boldsymbol{\tau}_h) + (e_u^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) - (e_p^{(n)}, \boldsymbol{\tau}_h \cdot \boldsymbol{n})_{\Gamma_S} = 0 \quad \forall \boldsymbol{\tau} \in V_h. \tag{A.7}$$

By (36b) and (A.1c),

$$\nabla \cdot \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} = 0 \quad (n \ge 1). \tag{A.8}$$

Substitute $\boldsymbol{\tau}_h = \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}$ into (A.7), we get

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_Q^2 - (e_p^{(n)}, \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n})_{\Gamma_S} = 0. \tag{A.9}$$

Since $P_{\tilde{\Lambda}_h}$ is contraction, subtracting (A.1a) from (58), we calculate as

$$\begin{aligned}
\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n}\|_\Lambda^2 &= \|P_{\tilde{\Lambda}_h}(\boldsymbol{\sigma} \cdot \boldsymbol{n} - \rho p_h) - P_{\tilde{\Lambda}_h}(\boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n} - \rho p_h^{(n-1)})\|_\Lambda^2 \\
&\le \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n} - \rho e_p^{(n-1)}\|_\Lambda^2 \\
&= \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n}\|_\Lambda^2 - 2\rho(e_p^{(n-1)}, \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n})_{\Gamma_S} + \rho^2 \|e_p^{(n-1)}\|_\Lambda^2 \\
&= \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n}\|_\Lambda^2 - 2\rho\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)}\|_V^2 + \rho^2\|e_p^{(n-1)}\|_\Lambda^2 \quad \text{(by (A.8) and (A.9))}.
\end{aligned} \tag{A.10}$$

Applying inf-sup condition (44) to (A.7), we have

$$\|e_u^{(n)}\|_Q \le C_4 \sup_{\boldsymbol{\tau}_h \in V_{h,\Gamma_S}} \frac{(e_u^{(n)}, \nabla \cdot \boldsymbol{\tau}_h)}{\|\boldsymbol{\tau}_h\|_V} = C_4 \sup_{\boldsymbol{\tau}_h \in V_{h,\Gamma_S}} \frac{-(\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}, \boldsymbol{\tau}_h) + 0}{\|\boldsymbol{\tau}_h\|_V} \le C_4 \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_Q. \tag{A.11}$$

To bound $e_p^{(n)}$, we apply the inf-sup condition (40) to (A.7) and derive

$$\begin{aligned}
\|e_p^{(n)}\|_\Lambda &\le C_3 \sup_{\boldsymbol{\tau}_h \in V_h} \frac{(e_p^{(n)}, \boldsymbol{\tau}_h \cdot \boldsymbol{n})_{\Gamma_S}}{\|\boldsymbol{\tau}_h\|_V} = C_3 \sup_{\boldsymbol{\tau}_h \in V_h} \frac{(\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}, \boldsymbol{\tau}_h) + (e_u^{(n)}, \nabla \cdot \boldsymbol{\tau}_h)}{\|\boldsymbol{\tau}_h\|_V} \\
&\le C_3(\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_Q + \|e_u^{(n)}\|_Q) \le C_3(1+C_4)\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_Q.
\end{aligned} \tag{A.12}$$

Hence, we have proved (A.6b) and (A.6c). It remains to validate (A.6a). Inserting (A.12) into the last term of (A.10), we get

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n}\|_\Lambda^2 \le \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n}\|_\Lambda^2 - (2\rho - \rho^2 C_3^2(1+C_4)^2)\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)}\|_Q^2. \tag{A.13}$$

Under the assumption (A.5), we see that $\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n}\|_\Lambda$ decreases by n unless $\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} = 0$. Therefore, $\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n}\|_\Lambda \downarrow 0$ as $n \to \infty$, which implies $(\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}, e_u^{(n)}) \to (\mathbf{0}, 0)$ because of (A.8) and

$$(\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}, \boldsymbol{\tau}_h) + (e_u, \nabla \cdot \boldsymbol{\tau}_h) = 0 \quad \forall \boldsymbol{\tau}_h \in V_{h,\Gamma_S}.$$

In addition, we have $e_p^{(n)} \to 0$ follows from (A.12). To show the convergence rate, we set $\alpha := 2\rho - \rho^2 C_3^2(1+C_4)^2$, and obtain from (A.3), (A.8) and (A.13) that

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n}\|_\Lambda^2 \le (1 - \alpha C_{T1}^{-2})\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n}\|_\Lambda^2 \le \cdots \le (1 - \alpha C_{T1}^{-2})^n \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(0)} \cdot \boldsymbol{n}\|_\Lambda^2. \tag{A.14}$$

One can take sufficiently small ρ to ensure that $1 - \alpha C_{T1}^{-2} < 1$. In fact, taking $\rho = (C_3^2(1+C_4)^2)^{-1}$ and in view of $C_3 C_{T1} \ge 1$ (by (A.4)), we have

$$\alpha C_{T1}^{-2} = (C_2 C_{T1})^{-2}(1+C_4)^{-2} \le (1+C_4)^{-2} < 1.$$

On the other hand, we have (by (A.4))

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V \le C_3 \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n}\|_\Lambda. \tag{A.15}$$

Note that (A.6a) is an immediate result from (A.14) and (A.15). We complete the proof. □

Remark A.2. Be aware that in order to ensure convergence (as shown in Eq. (A.13)), the value of ρ should be within the range of $(0, 2/(C_3^2(1+C_4)^2))$. However, it is important to note that selecting an excessively small value for ρ will slow down the convergence. When choosing $\rho = (C_3^2(1+C_4)^2)^{-1}$, α reaches the maximum value $(C_3^2(1+C_4)^2)^{-1}$. In this case, we obtain from (A.6a) that

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V^2 \le C_3^2(1 - C_{T1}^{-2}C_3^{-2}(1+C_4^{-2}))^n \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(0)}\|_V^2.$$

Remark A.3. As addressed in Remark A.1, the constant C_{T1} has an upper bound $Ch^{-\frac{1}{2}}$. Therefore, we have an upper bound of the iteration error:

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V^2 \le C(1 - Ch)^n \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(0)}\|_V^2,$$

which indicates, to attain $\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V = 10^{-k}$, it is necessary to perform a maximum of $n = O(2k/\log(1-Ch)) = O(kh^{-1})$ iterations. It reveals that the Uzawa algorithm may converge very slowly when taking a small mesh size h.

Algorithm 4 The Uzawa method based on **(MLM-II)**$_h$

(Step 1) Choose a $\rho > 0$ and $n = 1$. Take the same initial value $\boldsymbol{\sigma}_h^{(0)}$ and $p_h^{(0)}$ to the Algorithm 1.
(Step 2) Compute the projection

$$p_h^{(n)} = P_{\tilde{\Lambda}_h}(p_h^{(n-1)} - \rho \boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n}) = \max(0, p_h^{(n-1)} - \rho \boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n}). \tag{A.16}$$

(Step 3) Find $(\boldsymbol{\sigma}_h^{(n)}, u_h^{(n)}) \in V_h \times Q_h$ such that

$$(\boldsymbol{\sigma}_h^{(n)}, \boldsymbol{\tau}_h) + (u_h^{(n)}, \nabla \cdot \boldsymbol{\tau}_h) = (p_h^{(n)}, \boldsymbol{\tau}_h \cdot \boldsymbol{n})_{\Gamma_S} \qquad \forall \boldsymbol{\tau}_h \in V_h, \tag{A.17a}$$

$$(\nabla \cdot \boldsymbol{\sigma}_h^{(n)}, v_h) + (f_h, v_h) = 0 \qquad \forall v_h \in Q_h. \tag{A.17b}$$

(Step 4) $n \to n+1$. Iterate (Step2) (Step3) until convergence.

Proposition A.2. *Assume that ρ is sufficiently small such that*

$$0 < \rho < 2C_{T1}^{-2}. \tag{A.18}$$

Then $\|e_p^{(n)}\|_\Lambda \downarrow 0$ and $(\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}, e_u^{(n)}) \to (\boldsymbol{0}, 0)$ as $n \to \infty$. More precisely, we have $\nabla \cdot \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} = 0 \quad (\forall n)$, and

$$\|e_p^{(n)}\|_\Lambda^2 \le \left(1 - (2\rho - \rho^2 C_{T1}^2)C_3^{-2}(1+C_4)^{-2}\right)^n \|e_p^{(0)}\|_\Lambda^2, \tag{A.19a}$$

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V \le C_{T1}\|e_p^{(n)}\|_\Lambda, \tag{A.19b}$$

$$\|e_u^{(n)}\|_Q \le C_4 \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V. \tag{A.19c}$$

Proof. Similar to the proof of Proposition A.1, by subtracting (A.17a) from (37a), and taking $\boldsymbol{\tau}_h = \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}$, we get (A.7) and (A.9). It is apparent that (A.8), (A.11) and (A.12) also hold. Since $P_{\tilde{\Lambda}_h}$ is a contracting mapping, subtracting (A.16) from (64), we derive

$$\begin{aligned}
\|e_p^{(n)}\|_\Lambda^2 &= \|P_{\tilde{\Lambda}_h}(p_h - \rho \boldsymbol{\sigma}_h \cdot \boldsymbol{n}) - P_{\tilde{\Lambda}_h}(p_h^{(n-1)} - \rho \boldsymbol{\sigma}_h^{(n-1)} \cdot \boldsymbol{n})\|_\Lambda^2 \\
&\le \|e_p^{(n-1)} - \rho \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n}\|_\Lambda^2 \\
&= \|e_p^{(n-1)}\|_\Lambda^2 - 2\rho(e_p^{(n-1)}, \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n})_{\Gamma_S} + \rho^2 \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n}\|_\Lambda^2 \\
&= \|e_p^{(n-1)}\|_\Lambda^2 - 2\rho \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)}\|_V^2 + \rho^2 \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} \cdot \boldsymbol{n}\|_\Lambda^2 \quad \text{(by (A.9) and (A.8))} \\
&\le \|e_p^{(n-1)}\|_\Lambda^2 - (2\rho - \rho^2 C_{T1}^2)\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)}\|_V^2 \quad \text{(by (A.3))}.
\end{aligned} \tag{A.20}$$

Under the assumption (A.18), i.e., $\alpha := 2\rho - \rho^2 C_{T1}^2 > 0$, we find that $\|e_p^{(n)}\|_\Lambda$ decreases by n unless $\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n-1)} = \boldsymbol{0}$. Hence, we have $\|e_p^{(n)}\|_\Lambda \downarrow 0$, which implies $\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \to \boldsymbol{0}$ and $e_u^{(n)} \to 0$ $(n \to \infty)$. Applying (A.12) to the last term of (A.20), we obtain

$$\begin{aligned}
\|e_p^{(n)}\|_\Lambda^2 &\le (1 - \alpha C_3^{-2}(1+C_4)^{-2})\|e_p^{(n-1)}\|_\Lambda^2 \\
&\le \cdots \le (1 - \alpha C_3^{-2}(1+C_4)^{-2})^n \|e_p^{(0)}\|_\Lambda^2.
\end{aligned} \tag{A.21}$$

Note that $1 - \alpha C_3^{-2}(1+C_4)^{-2} > 0$ can be guaranteed by taking sufficiently small ρ. In fact, choosing $\rho = C_{T1}^{-2}$, we have $\alpha = 2\rho - \rho^2 C_{T1}^2 = C_{T1}^{-2}$. Moreover, in view of $C_3 C_{T1} \ge 1$ (by (A.4)), we have

$$\alpha C_3^{-2}(1+C_4)^{-2} = (C_{T1}C_3)^{-2}(1+C_4)^{-2} \le (1+C_4)^{-2} < 1.$$

It follows from (A.9) and (A.8) that

$$\|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V^2 = \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_Q^2 = (e_p^{(n)}, \boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n})_{\Gamma_S} \le \|e_p^{(n)}\|_\Lambda \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)} \cdot \boldsymbol{n}\|_\Lambda \le \|e_p^{(n)}\|_\Lambda C_{T1} \|\boldsymbol{e}_{\boldsymbol{\sigma}}^{(n)}\|_V,$$

which is $\|e_{\sigma}^{(n)}\|_V \le C_{T1}\|e_p^{(n)}\|_\Lambda$. Together with (A.21) and (A.11), we conclude (A.19). □

Remark A.4. As with Remarks A.2 and A.3, we give a rough calculation of the number of iterations to reach the tolerance error. By choosing $\rho = C_{T1}^{-2}$, it follows from (A.19a) and (A.19b) that

$$\|e_{\sigma}^{(n)}\|_V^2 \le (C_{T1}^2(1 - C_{T1}C_3)^{-2}(1 + C_4)^{-2})^n \|e_p^{(0)}\|_\Lambda^2.$$

Recalling that C_{T1} has an upper bound $Ch^{-\frac{1}{2}}$ (see Remark A.1), we find that $\|e_{\sigma}^{(n)}\|_V^2 \le C(h^{-1}(1 - Ch)^n)\|e_p^{(0)}\|_\Lambda^2$. Therefore, it is necessary to perform a maximum of $n = O((k + |\log h|)h^{-1})$ iterations to attain $\|e_{\sigma}^{(n)}\|_V \le 10^{-k}$, which suggests that the Uzawa method is inefficient.

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